

5 Choice under Uncertainty

Thus far we have analyzed preference relations, utility functions, and utility maximization in deterministic settings. In the preceding chapters, the decision maker faced no uncertainty about the outcomes of his actions. However, we know that our daily choices are often surrounded by uncertainty: Will my pen stop working in the middle of a test? Will I find a traffic jam on my way to work? In our analysis of decision making under uncertainty, we start by considering simple lotteries, understood as a list of potential outcomes, each outcome with an associated probability of occurrence. We then analyze “lotteries of lotteries” (referred to as compound lotteries), where the decision maker faces a list of lotteries and plays each lottery with a specific probability. While this description of lotteries is useful in our representation of uncertainty, our ultimate goal is to describe individual’s preferences over lotteries, as we elaborate in section 5.1. In particular, we seek to identify which properties are required on preference relations over lotteries to guarantee that they can be represented by a utility function.¹ We now require a new property that did not arise in deterministic contexts: the “independence axiom.” Essentially, this axiom states that if a decision maker prefers one lottery over another, then introducing a small perturbation on the probabilities of the two initial lotteries does not entail a change in his preferences. In section 5.2 we examine this axiom and the controlled experiments in which participants violated the axiom. We also discuss the most important theoretical developments that, mainly in the last three decades, have been offered to account for this type of behavior. In section 5.3 we focus on a specific type of lottery for which outcomes are restricted to be monetary amounts,

¹ Until now we have learned that in deterministic settings we only need preferences to satisfy monotonicity and continuity (or in alternative approaches, completeness, transitivity, local nonsatiation, and continuity) for them to be representable by a utility function. Most of these properties are also required in preferences over lotteries for them to be representable by a utility function, and in particular, completeness, transitivity, and continuity.

rather than any type of N -dimensional bundle. Section 5.4 then presents behavioral theories that modify or expand standard expected utility theory, such as “weighted utility theory” or “rank-dependent utility” and their properties.

Section 5.5 examines how to measure an individual’s risk aversion by evaluating the curvature of his utility function, and discusses the Arrow–Pratt coefficient of absolute and relative risk aversion. Then section 5.6 compares payoff distributions in money lotteries, by which a decision maker chooses the lottery with the largest expected value (as captured by the notion of first-order stochastic dominance), and if two lotteries have the same mean, he chooses the lottery with the smallest variance (as represented by the concept of second-order stochastic dominance). We also present other common measures of stochastic dominance often used in the literature. Section 5.7 analyzes reference-dependent utility and prudence, section 5.8 examines prospect theory, and section 5.9 discusses common measures to compare payoff distributions. Section 5.10 deals with subjective probability theory, whereby the probability of each event is not objectively observed but is instead inferred from individual choices, and thus prone to subjectivities. Section 5.11 explores three alternative theories to subjective probability: ambiguity aversion, smooth ambiguity aversion, and the theory of capacities (also known as Choquet expected utility). Last, the chapter provides an appendix that covers state-dependent utility, where the utility that the decision maker obtains in each state can differ.

5.1 Simple and Compound Lotteries

In contrast to our analysis in previous chapters where the individual or firm selects among a set of certain outcomes, we now examine choices under uncertain conditions. In this section we present the decision maker’s preferences over uncertain outcomes, and discuss how to represent his preference relation with an expected utility function.

In particular, consider a set of possible outcomes (or consequences) C . This set might include, for instance, simple monetary payoffs (either positive or negative), implying that the set of consequences becomes $C = \mathbb{R}$. Alternatively, set could represent consumption bundles, in which case $C = X$ (where X is a subset of $\mathbb{R}^L$, as in previous chapters). For simplicity, assume that there are only a finite number of outcomes; that is, the set of possible outcomes C contains N elements. Also the probabilities associated to every possible outcome are objectively known,² being p_1 for outcome 1, p_2 for outcome 2, and so on.

² In section 5.11, we allow for situations where the decision maker does not perfectly know the probability associated to every outcome.

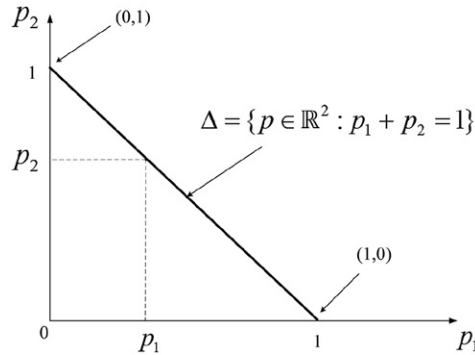


Figure 5.1
Lottery over two outcomes

5.1.1 Simple Lotteries

In this chapter we use the concept of lotteries to represent uncertain outcomes. In particular, a simple lottery is a list

$$L = (p_1, p_2, \dots, p_N),$$

where p_n is the probability of outcome n occurring, and satisfies $p_n \geq 0$ for every outcome n , and $\sum_{n=1}^N p_n = 1$. Therefore we can graphically represent a simple lottery with two possible outcomes as a point along the line connecting the probability point $(0, 1)$, where outcome 2 happens with certainty, and $(1, 0)$, where outcome 1 occurs with certainty, as figure 5.1 depicts. This is known as the “simplex” of lotteries, Δ .

Intuitively, the horizontal (vertical) intercept represents a “degenerated” probability distribution, where outcome 1 (outcome 2, respectively) is certain. Strictly positive probability pairs (p_1, p_2) along the line $p_1 + p_2 = 1$, in contrast, describe a lottery where none of the outcomes is certain, and therefore the individual faces some uncertainty. We can easily extend this graphical representation of lotteries to the case of three possible outcomes with associated probabilities (p_1, p_2, p_3) , as figure 5.2 illustrates. First, note that the intercepts also represent degenerated probabilities where one outcome is certain. Second, points strictly inside the hyperplane connecting the three intercepts denote a lottery where the individual faces uncertainty, such as the point depicted in figure 5.2. This figure is usually referred as the probability simplex of lotteries with $N = 3$ outcomes.

In order to simplify our graphical analysis, we can do a two-dimensional projection of the hyperplane above, as figure 5.3 depicts. First, the vertices represent the intercepts (where one outcome is certain). Second, a simple lottery (interior points in the

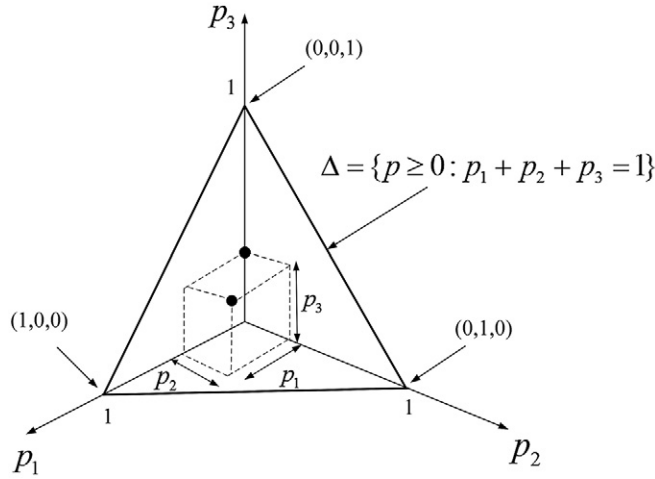


Figure 5.2
Simplex of a lottery with three outcomes

triangle) represents settings where the individual faces uncertainty. Importantly, the distance from a given point to the side of the triangle measures the probability that the outcome represented at the opposite vertex occurs. In figure 5.3, for instance, the length of the segment x_1 (from the strictly interior point to the side of the triangle connecting the certain outcomes 2 and 3) measures the probability of outcome 1 occurring.

Finally, as a practice, if a lottery lies on one of the boundaries of the triangle, then we can only construct segments connecting the lottery to two of the outcomes, entailing that the probability associated with the third outcome is zero. Figure 5.4 illustrates this case.

5.1.2 Compound Lotteries

We can use our previous notation to define compound lotteries. Specifically, we consider a list of K simple lotteries, where

$$L_k = (p_1^k, p_2^k, \dots, p_K^k) \quad \text{for every lottery } k = 1, 2, \dots, K,$$

with associated probabilities $\alpha_k \geq 0$ for every lottery k , where $\sum_{k=1}^K \alpha_k = 1$. Then the compound lottery $(L_1, L_2, \dots, L_K; \alpha_1, \alpha_2, \dots, \alpha_K)$ yields the simple lottery L_k with probability α_k .

We can intuitively interpret a compound lottery as a “lottery of lotteries.” First, we face a probability α_1 of playing lottery L_1 , and if lottery 1 occurs, then we face a

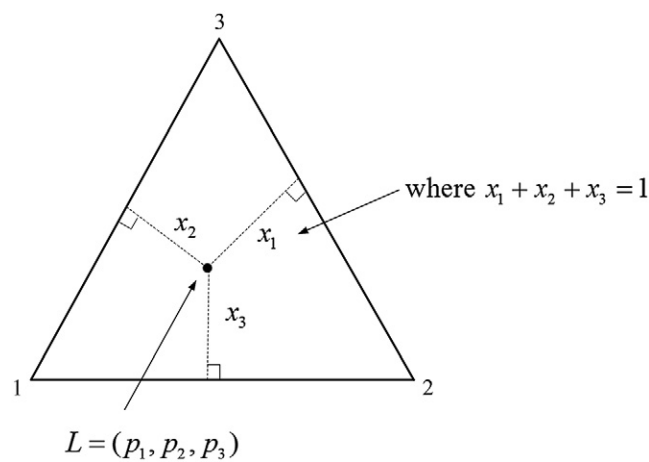


Figure 5.3
Two-dimensional representation of a simplex

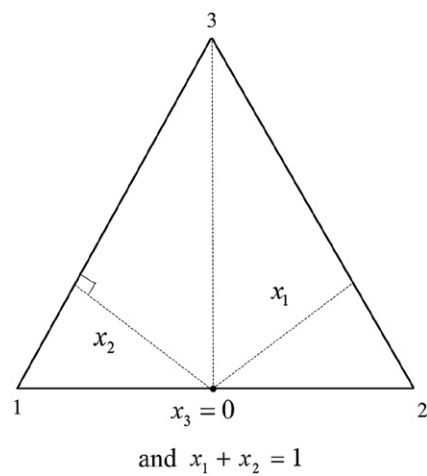


Figure 5.4
Lottery assigning zero probability to outcome 3

probability p_1^1 of outcome 1 occurring, probability p_1^2 of outcome 2 occurring, and so on. In this context, the probability of outcome 1 is given by the probability that we play lottery L_1 and outcome 1 occurs, $\alpha_1 \cdot p_1^1$, plus the probability that such outcome occurs in lottery L_2 , $\alpha_2 \cdot p_1^2$, and similarly across all K lotteries. (Recall that p_j^k represents the probability that outcome j occurs when playing lottery k .) That is, we can write

$$p_1 = \alpha_1 \cdot p_1^1 + \alpha_2 \cdot p_1^2 + \dots + \alpha_K \cdot p_1^K,$$

which describes the joint probability of outcome 1 occurring in lottery 1, 2, ..., K . Extending the same argument to all possible outcomes in the compound lottery $(L_1, L_2, \dots, L_K; \alpha_1, \alpha_2, \dots, \alpha_K)$, we can calculate a corresponding reduced lottery as the simple lottery $L = (p_1, p_2, \dots, p_N)$ that generates the same ultimate probability distribution of outcomes as the compound lottery above. That is, the reduced lottery L of any compound lottery can be obtained by

$$L = \alpha_1 L_1 + \alpha_2 L_2 + \dots + \alpha_K L_K \in \Delta.$$

Example 5.1: Compound lotteries Let us consider two examples of reduced lotteries. In figure 5.5 all lotteries are equally likely (the probability of each lottery occurring is $\alpha_i = 1/3$ for every lottery $i = 1, 2, 3$). If lottery 1 occurs, we are guaranteed outcome 1, but if lottery 2 or 3 occurs, we face a positive probability of obtaining any of the three possible outcomes. The probability of outcome 1 in this compound lottery is therefore

$$p_1 = \frac{1}{3}1 + \frac{1}{3}\frac{1}{4} + \frac{1}{3}\frac{1}{4} = \frac{1}{2}.$$

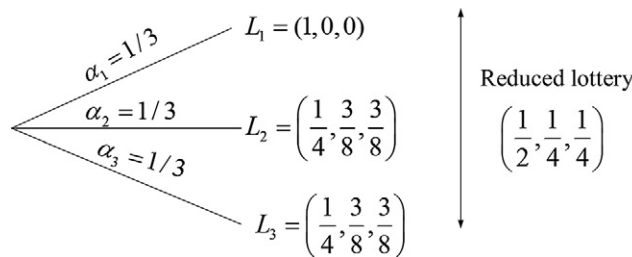


Figure 5.5
Compound lottery and its associated reduced lottery (example 1)

Similarly the probability of outcome 2 is

$$p_2 = \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot \frac{3}{8} + \frac{1}{3} \cdot \frac{3}{8} = \frac{1}{4}.$$

(The probability of outcome 3 can be found in a similar manner, also being $1/4$.) The reduced lottery of the compound lottery represented in Figure 5.5 is therefore $(1/2, 1/4, 1/4)$. Figure 5.6a locates the three simple lotteries L_1 , L_2 , and L_3 in the probability simplex, and also depicts the reduced lottery L that assigns the same probability weight to each lottery. ■

Example 5.2: Finding the reduced lottery of a compound lottery Let us consider another example (see figure 5.6b). Lotteries 4 and 5 are equally likely, meaning $\alpha_4 = \alpha_5 = 1/2$. While outcome 1 occurs with 50 percent probability on both lotteries, lottery 4 assigns the remaining probability weight to outcome 2, and lottery 5 assigns the same probability weight to outcome 3. The figure also describes the joint probability of outcomes 1, 2, and 3 occurring in the compound lottery, which helps us identify the three probability weights associated to every outcome in the reduced lottery.

The reduced lottery is therefore $(1/2, 1/4, 1/4)$, which is represented as a convex combination between lotteries L_4 and L_5 in the simplex of figure 5.7. Interestingly, both compound lotteries in examples 5.1 and 5.2 induce the same reduced lottery, despite originating from a different set of simple lotteries. ■

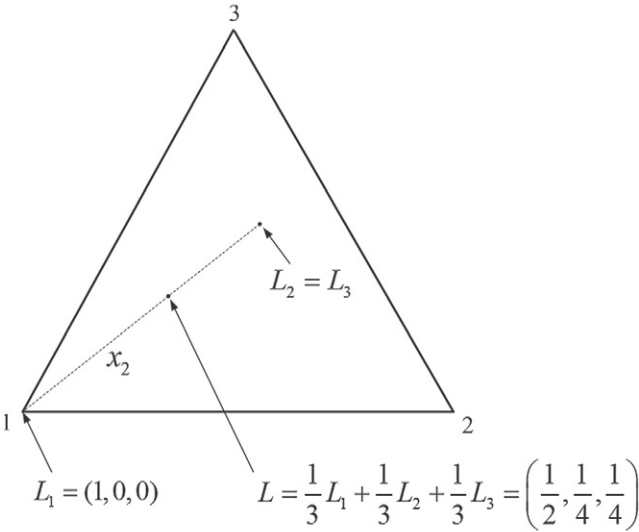
5.2 Preferences over Lotteries

Let us turn to the preferences of decision makers who face uncertain outcomes. We assume that individuals only care about the compound lotteries that induce a given reduced lottery, as in the previous example, where two different compound lotteries produce the same reduced lottery. We refer to this assumption as “consequentialism,” since only consequences (outcomes), and the probability associated with every consequence, matter for the decision maker, rather than the process by which a given outcome emerges.

In addition we consider the set of all simple lotteries over the set of outcomes, $\mathcal{L}$. We assume that the decision maker has a complete and transitive preference relation over lotteries in $\mathcal{L}$, allowing him to compare any pair of simple lotteries L and L' . That is,

1. *Completeness* Either $L \succsim L'$ or $L' \succsim L$, or both, for every two lotteries $L, L' \in \mathcal{L}$.
2. *Transitivity* If $L \succsim L'$ and $L' \succsim L''$, then $L \succsim L''$ for every three lotteries $L, L', L'' \in \mathcal{L}$.

a.



b.

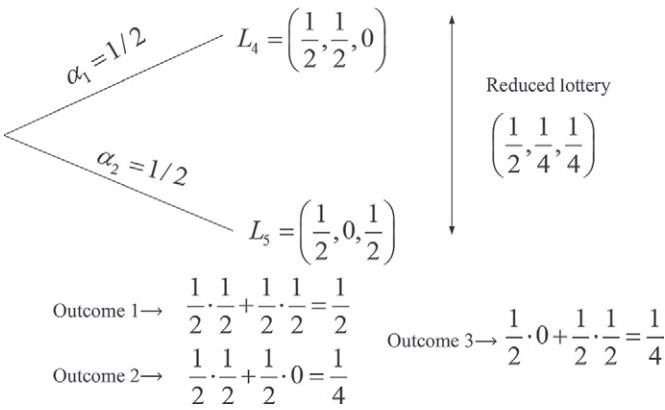


Figure 5.6
(a) Simplex of the reduced lottery of a compound lottery (example 1). (b) Reduced lottery of a compound lottery (example 2)

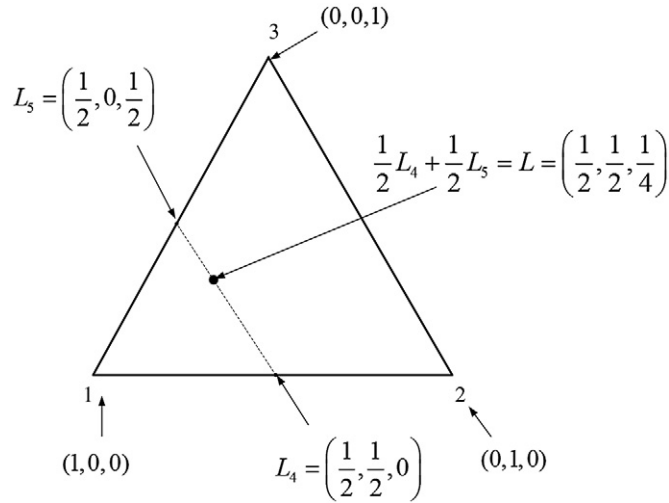


Figure 5.7
Simplex of reduced lottery (example 2)

Example 5.3: Four examples of preferences over lotteries Let us first consider examples of preference relations over lotteries where the decision maker is only concerned about the probability distribution over outcomes.

1. *Extreme preference for certainty* The decision maker weakly prefers lottery L to L' , meaning $L \succeq L'$, if and only if

$$\max_{n \in N} p_n \geq \max_{n \in N} p'_n.$$

Intuitively, this preference relation represents a decision maker who is only concerned about the probability associated with the most likely outcome. That is, he first considers the most likely outcome in lottery L and that in lottery L' , and afterward he chooses the lottery in which such outcome is the most likely.³ For example, a decision maker with these preferences would prefer lottery $L = (70\%, 30\%)$, which assigns a probability of 70 percent to the outcome \$1 and 30 percent to an outcome of \$100, to lottery $L' = (40\%, 60\%)$, which assigns a probability of 40 percent to \$1, and a probability of 60 percent to \$100. Specifically, $\max\{p_n : p_n \geq 0\} = 70$ percent for lottery L , but only 60 percent for lottery L' .

³ You might consider this preference relation to be reasonable. Note, however, that the most likely outcome might differ from lottery L to lottery L' , making this preference relation over uncertain outcomes not so sensible in certain settings.

2. *Smallest size of the support* For two lotteries L and L' over a finite number of actions, the decision maker weakly prefers lottery L to L' , since $L \succeq L'$, if and only if

$$|\text{supp}(L)| \leq |\text{supp}(L')|,$$

where $|\text{supp}(L)|$ denotes the cardinality of the support of lottery L , namely the number of outcomes receiving a strictly positive probability, where $\text{supp}(L) = \{n \in N: p_n > 0\}$. Intuitively, this decision maker prefers the lottery whose probability distribution is concentrated over the smallest set of possible outcomes. Note that this preference relation only cares about the probability distribution. For example, an individual with this preferences would prefer a lottery L that assigns the same probability to \$5 and \$6 (a very small support of possible outcomes) than a lottery L' assigning the same probability to all outcomes in a longer list of monetary outcomes that starts at \$10, that is, in the list $\{\$10, \$11, \dots, \$100\}$. In particular, $|\text{supp}(L)|=2$ while $|\text{supp}(L')|=90$, entailing that $L \succeq L'$ for a decision maker with this type of preference over lotteries.

Let us next examine preference relations over lotteries for which the decision maker cares not only about probability distributions but also about the lottery outcomes.

3. *Lexicographic preferences* After ordering outcomes from most to least preferred, the decision maker weakly prefers lottery L to L' so that $L \succeq L'$, if and only if

$$p_1 > p'_1 \quad \text{or}$$

$$p_1 = p'_1 \quad \text{and} \quad p_2 > p'_2, \quad \text{or}$$

$$p_1 = p'_1, \quad p_2 = p'_2, \quad \text{and} \quad p_3 > p'_3, \quad \text{and so on.}$$

Intuitively, the decision maker weakly prefers lottery L to L' if outcome 1 (the most preferred outcome) is more likely to occur in lottery L than in lottery L' . If such outcome is equally likely in both lotteries, meaning $p_1 = p'_1$, then the decision maker prefers lottery L to L' if outcome 2 (the second most preferred outcome) is more likely to occur in lottery L than in lottery L' , meaning $p_2 > p'_2$. As an example, for monetary outcomes (\$3, \$2, \$-100) a decision maker with a lexicographic preference relation would prefer a lottery $L = (2\%, 0\%, 98\%)$ over $L' = (1\%, 99\%, 0\%)$, since $p_1 = 2\% > 1\% = p'_1$, despite of the fact that L assigns a large probability to the lowest payoff \$-100.

4. *Worst case scenario* First, the decision maker attaches a real number to every outcome $z \in C$, $v(z) \in \mathbb{R}$, that can be interpreted as the utility from outcome z . Then, he weakly prefers lottery L to L' , that is, $L \succeq L'$, if and only if

$$\min\{v(z): p(z) > 0\} > \min\{v(z): p'(z) > 0\}.$$

Intuitively, this decision maker prefers lottery L if the lowest utility he can get from playing lottery L , $\min v(z)$, is higher than the lowest utility he can obtain from playing lottery L' , among all outcomes that can occur with a positive probability. For this reason this preference relation is often regarded as that the decision maker first identifies the “worst case scenario” of each lottery, and then prefers the lottery with the *best* “worst case scenario.”⁴ ■

Let us next define continuity of preferences in this context. For completeness, we present two equivalent definitions.

Continuity 1 For any three lotteries L , L' , and L'' , sets

$$\{\alpha \in [0, 1]: \alpha L + (1 - \alpha)L' \succsim L''\} \quad \text{and} \quad \{\alpha \in [0, 1]: L'' \succsim \alpha L + (1 - \alpha)L'\}$$

are closed.

In words, for a lottery L'' , we can construct compound lotteries between L and L' that are weakly preferred to L'' (in the first set). Such a set of compound lotteries is closed because it contains all its limiting points. That is, it contains the “limiting compound lotteries” that the individual regards as indifferent to L'' . A similar argument applies to the set of compound lotteries for which the individual weakly prefers L'' to $\alpha L + (1 - \alpha)L'$.

The following definition of continuity is perhaps more intuitive. We elaborate on its interpretation below.

Continuity 2 If lottery L is strictly preferred to L' , $L \succ L'$, then there is a small neighborhood around L and around L' , $B(L)$ and $B(L')$, respectively, such that for all lotteries $L_a \in B(L)$ and $L_b \in B(L')$, we have that $L_a \succ L_b$.

Figure 5.8 illustrates the intuition behind this definition. In particular, small changes in the probability distribution of lotteries L and L' do not change the decision maker’s preference over the two lotteries.

To better see this, let us consider an example from Mas-Colell et al. (1995). If a decision maker prefers a car trip to staying at home (both events occurring with certainty), then he must still prefer the car trip (if we introduce a small probability of suffering a car

⁴ This preference ordering is sometimes observed in computer science, where one algorithm is preferred to another if it works better in the worst case scenario, independently of the probability that such worst case scenario occurs (as long as it is positive).

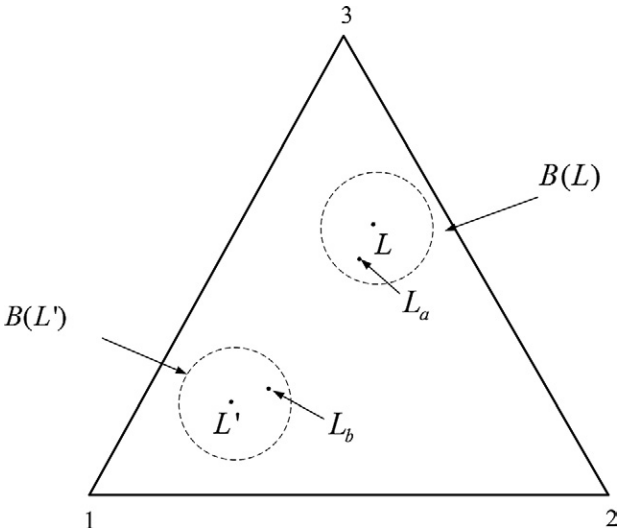


Figure 5.8
Continuity of preference over lotteries

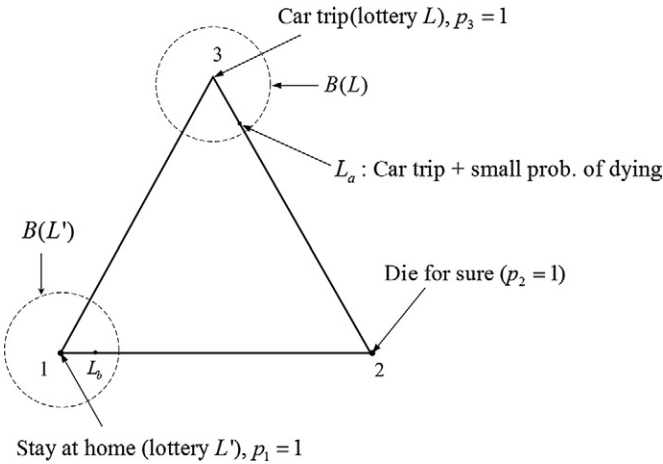


Figure 5.9
Continuity of preferences in the car trip example

accident) than staying at home. As figure 5.9 depicts, he slightly moves from the vertex representing the certain event “car trip” (at the top of figure 5.9) to lottery L_a (which represents a car trip with a small probability of a car accident) but still prefers lottery L_a to lottery L' (staying at home), or any lottery in its neighborhood, such as L_b .

The continuity assumption above, as in consumer theory, implies the existence of a utility function from the set of all lotteries in $\mathcal{L}$ to the real numbers, $U: \mathcal{L} \rightarrow \mathbb{R}$, such that lottery L is weakly preferred to lottery L' , $L \succeq L'$, if and only if $U(L) \geq U(L')$.

We must, however, impose an additional assumption on preferences over lotteries in order to guarantee that the decision maker’s preferences satisfy “consequentialism” as suggested above. We do so by imposing the following independence axiom (IA).

Independence axiom (IA) A preference relation over lotteries satisfies the IA if, for any three lotteries L , L' , and L'' , and for $\alpha \in (0, 1)$, we have that $L \succeq L'$ if and only if the compound lottery $\alpha L + (1 - \alpha)L''$ satisfies

$$\alpha L + (1 - \alpha)L'' \succeq \alpha L' + (1 - \alpha)L''.$$

Intuitively, if we mix each of two lotteries, L and L' , with a third one, L'' , the preference ordering of the two resulting compound lotteries only depends on the preference ordering of the two simple lotteries, L and L' . The preference ordering of the two resulting compound lotteries does not depend on (is independent of) the particular third lottery L'' that we use. We provide a graphical illustration of the IA below. Specifically,

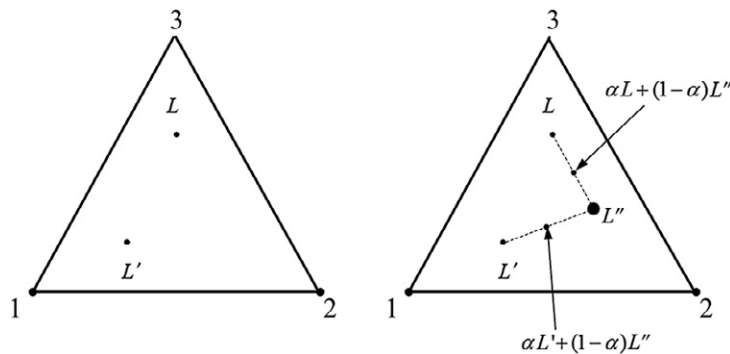


Figure 5.10
Independence axiom (IA)

in the left panel of figure 5.10 the individual prefers lottery L to L' . Hence it must be that, when we construct a convex combination of the first two lotteries with any third lottery L'' (right-hand panel of figure 5.10), the convex combination of L and L'' , $\alpha L + (1 - \alpha)L''$, is still preferred to that of L' and L'' , $\alpha L' + (1 - \alpha)L''$, meaning

$$\alpha L + (1 - \alpha)L'' \succeq \alpha L' + (1 - \alpha)L''.$$

The following example illustrates the intuition behind the IA. Consider a decision maker who prefers lottery L to L' . We can construct a compound lottery where, after a coin toss, the decision maker plays lottery L when heads comes up and L'' when tails does, and another compound lottery where the decision maker plays lottery L' when heads comes up and L'' when tails does. The IA tells us that this decision maker, who preferred lottery L to L' , must still prefer the first to the second compound lottery.⁵

Example 5.4: Testing the IA It is straightforward to show that not all preference relations satisfy the IA. For instance, the “extreme preference for certainty” preferences we described in example 5.3 violate the IA. In particular, starting from two simple lotteries L and L' for which $L \sim L'$, we can construct two compound lotteries for which $\frac{1}{2}L + \frac{1}{2}L \neq \frac{1}{2}L' + \frac{1}{2}L$. Intuitively, if the individual is indifferent between lotteries L and L' , it must be because the most likely outcome in L and L' occurs with exactly the same probability in both lotteries, that is,

$$\max_{n \in N} \{p_1, p_2, \dots, p_n\} = \max_{n \in N} \{p'_1, p'_2, \dots, p'_n\}.$$

The compound lottery $\frac{1}{2}L + \frac{1}{2}L$ coincides with the simple lottery L , and therefore the individual still judges it according to $\max_{n \in N} \{p_1, p_2, \dots, p_n\}$. In contrast, the compound lottery $\frac{1}{2}L' + \frac{1}{2}L$ implies a reduced lottery with associated probabilities

$$\{\frac{1}{2}p'_1 + \frac{1}{2}p_1, \dots, \frac{1}{2}p'_n + \frac{1}{2}p_n\}$$

⁵ While the IA seems a sensible assumption in the theory of choice under uncertainty, note that it does not necessarily hold in consumer theory when outcomes are certain. In particular, a consumer might prefer good A over good B , but the combination of A with a third good C does not need to be preferred to the combination of B with the third good C . This preference ordering might occur when, for instance, the consumer regards A and C as substitutes but B and C as complements in consumption, thereby making the combination of B and C more attractive than that of A and B .

whose maximum does not necessarily coincide with the maximum of lottery L ; that is, $\max_{n \in N} \{\frac{1}{2}p'_1 + \frac{1}{2}p_1, \dots, \frac{1}{2}p'_n + \frac{1}{2}p_n\}$ might differ from $\max_{n \in N} \{p'_1, p'_2, \dots, p'_n\}$. Indeed let us take a numerical example. Consider lotteries $L = (0.4, 0.5, 0.1)$ and $L' = (0.5, 0, 0.5)$. Here $\max\{0.4, 0.5, 0.1\} = 0.5$ implies that the decision maker is indifferent between both lotteries, $L \sim L'$. However, the compound lottery $\frac{1}{2}L + \frac{1}{2}L'$ entails probabilities

$$\left(\frac{0.4+0.5}{2}, \frac{0.5+0}{2}, \frac{0.1+0.5}{2} \right) = (0.45, 0.25, 0.3).$$

Therefore $\max\{0.45, 0.25, 0.3\} = 0.45$, implying that $\max\{0.4, 0.5, 0.1\} = 0.5 > \max\{0.45, 0.25, 0.3\} = 0.45$, which entails that $\frac{1}{2}L + \frac{1}{2}L \succ \frac{1}{2}L + \frac{1}{2}L'$, thus violating the IA (which would require $\frac{1}{2}L + \frac{1}{2}L' \sim \frac{1}{2}L + \frac{1}{2}L$).

Given these assumptions, we can now state that the utility function over lotteries has the so-called expected utility form.

Expected utility The utility function $U: \mathcal{L} \rightarrow \mathbb{R}$ has the expected utility form if there is an assignment of numbers $(u_1, u_2, \dots, u_N)$ to the N possible outcomes such that, for every simple lottery $L = (p_1, p_2, \dots, p_N) \in \mathcal{L}$, we have

$$U(L) = p_1u_1 + p_2u_2 + \dots + p_Nu_N.$$

A utility function with the expected utility form is also referred as a von Neumann–Morgenstern (vNM) expected utility function. Note that this function is linear in the probabilities.⁶ In particular,

$$U\left(\sum_{k=1}^K \alpha_k L_k\right) = \sum_{k=1}^K \alpha_k \cdot U(L_k)$$

for any lottery $L_k \in \mathcal{L}$, $k = 1, 2, \dots, K$ with associated probability $\alpha_k > 0$. Intuitively, the utility of the expected value of K lotteries, $U\left(\sum_{k=1}^K \alpha_k L_k\right)$, coincides with the expected

⁶ For more examples of utility functions over lotteries that violate the vNM expected utility representation, see exercise 3 at the end of the chapter. For a proof of the expected utility theorem, namely the existence of an expected utility representation that is linear in the probabilities, see Mas-Colell et al. (1995, 176–79).

utility of K lotteries, $\sum_{k=1}^K \alpha_k \cdot U(L_k)$. Since probabilities enter linearly, we can write the utility of the expected value of playing the K lotteries (left-hand side of the above expression) as

$$U\left(\sum_{k=1}^K \alpha_k L_k\right) = \sum_{n=1}^N u_n \left(\sum_{k=1}^K \alpha_k p_n^k\right),$$

where, for a given outcome n , the decision maker finds the joint probability of outcome n occurring in lottery 1, $\alpha_1 p_n^1$, plus the joint probability of the same outcome n occurring in lottery 2, $\alpha_2 p_n^2$, and similarly for all K lotteries. Summing the joint probability of outcome n occurring across the K lotteries, we obtain the total joint probability of outcome n occurring $\sum_{k=1}^K \alpha_k p_n^k$, which we multiply by the utility that the decision maker derives from outcome n , u_n . We can then repeat this process for every possible outcome $n = 1, 2, \dots, N$. Similarly, the expected utility from playing the K lotteries (right-hand side of the above expression) is represented by

$$\sum_{k=1}^K \alpha_k U(L_k) = \sum_{k=1}^K \alpha_k \left(\sum_{n=1}^N u_n p_n^k\right),$$

where, for a given lottery k , we find the expected utility from outcome 1 occurring in lottery k , $u_1 p_1^k$, plus the expected utility from outcome 2 occurring in the same lottery k , $u_2 p_2^k$, and so on. If we sum over all possible outcomes, we will obtain the expected utility from playing a given lottery k , $\sum_{n=1}^N u_n p_n^k$. We can multiply this expected utility by the associated probability of lottery k occurring, α_k , and then repeat this process for all lotteries $k = 1, 2, \dots, K$.

Cardinality The expected utility property presented above is a cardinal property. That is, not only the ranking matters but also the particular number resulting from the expected utility function $U: \mathcal{L} \rightarrow \mathbb{R}$ matters. Hence the expected utility form of an initial utility function $U(L)$ is preserved only under increasing linear transformations (also referred to as affine transformations), such as $\beta U(L) + \gamma$, where $\beta > 0$ and $\gamma > 0$, as the following result confirms.⁷

Representability Suppose that a decision maker's preference relation over lotteries satisfies rationality (completeness and transitivity), continuity, and the independence

⁷ This property contrasts with that of the theory of demand under certainty, where cardinality was not required; ordinality was sufficient. In that context we could apply a nonlinear transformation $g(\cdot)$ to utility function $u(\cdot)$ and still guarantee that $g(u(\cdot))$ represents the same preference relation as $u(\cdot)$ does.

axiom. This preference relation admits a utility representation of the expected utility form. That is, we can assign a number u_n to every outcome $n = 1, 2, \dots, N$ in such a manner that, for any two lotteries $L = (p_1, p_2, \dots, p_n)$ and $L' = (p'_1, p'_2, \dots, p'_n)$, lottery L is weakly preferred to lottery L' if and only if $U(L) \geq U(L')$, or $\sum_{n=1}^N p_n \cdot u_n \geq \sum_{n=1}^N p'_n \cdot u_n$. (Here u_n is the utility that the decision maker assigns to outcome n , and it is usually referred to as the Bernoulli utility function.)

Up to this point the decision maker's preference over lotteries had not been graphically represented with indifference curves. Let us next analyze the effect of the IA on indifference curves over lotteries. In particular, the IA implies that indifference curves must be straight and parallel lines. Let us separately analyze each of these properties.

1. *Indifference curves must be straight lines.* If a decision maker is indifferent between two lotteries L and L' , then the IA implies that he must be indifferent between the compound lotteries $\alpha L + (1 - \alpha)L$ and $\alpha L + (1 - \alpha)L'$ for all $0 < \alpha < 1$ (note that we only added αL on both sides of the indifference relation and hence $\alpha L + (1 - \alpha)L = L$). This result is graphically illustrated in figure 5.11, where the decision maker is indifferent between lotteries L and L' , and therefore he must also be indifferent between L and any convex combination of L and L' , as graphically represented by the line connecting lotteries L and L' .

Alternatively, a decision maker may be indifferent between lotteries L and L' . Then, using the IA, we would obtain that he must be indifferent between $\frac{1}{2}L + \frac{1}{2}L$ and $\frac{1}{2}L' + \frac{1}{2}L$. This is graphically represented in figure 5.12, where the individual is indifferent between lotteries L and L' (they both lie on the same indifference curve). In this setting the IA implies that the compound lottery $\frac{1}{2}L' + \frac{1}{2}L$ should

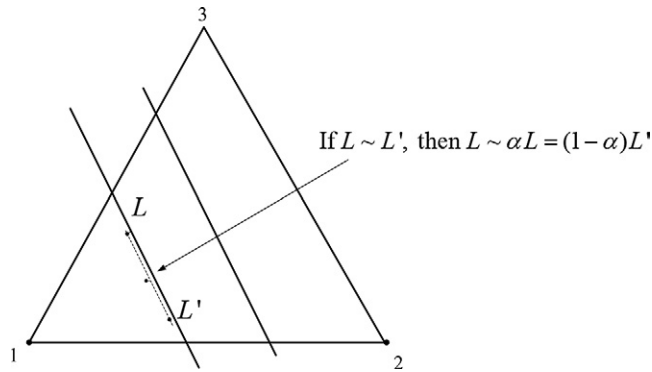


Figure 5.11
Effects of the IA whereby indifference curves are straight lines

also lie on the same indifference curve as $\frac{1}{2}L + \frac{1}{2}L$, which coincides with the simple lottery L , graphically entailing that the indifference curves be straight. Note that if instead the indifference curves were curvy—as depicted in figure 5.12—the compound lottery $\frac{1}{2}L' + \frac{1}{2}L$ would not lie on the same indifference curve as lotteries L and L' , and hence the decision maker would violate the IA. That is to say, despite being indifferent between simple lotteries L and L' , he is *not* indifferent between the compound lotteries $\frac{1}{2}L + \frac{1}{2}L$ and $\frac{1}{2}L' + \frac{1}{2}L$.

2. *Indifference curves must be parallel.* If a decision maker is indifferent between two lotteries L and L' , then the IA implies that he must be indifferent between the compound lotteries $\frac{1}{3}L + \frac{2}{3}L''$ and $\frac{1}{3}L' + \frac{2}{3}L''$. This is depicted in figure 5.13 where, starting from two lotteries L and L' over which the decision maker is indifferent (and therefore they lie on the same indifference curve), the convex combination of each of these two lotteries with a third lottery L'' also lies on the same indifference curve. If instead compound lotteries $\frac{1}{3}L + \frac{2}{3}L''$ and $\frac{1}{3}L' + \frac{2}{3}L''$ were on distinct indifference curves—as they do in figure 5.13 where indifference curves are not parallel—then the IA is violated. Notice that, in order for the decision maker to be indifferent between the compound lotteries $\frac{1}{3}L + \frac{2}{3}L''$ and $\frac{1}{3}L' + \frac{2}{3}L''$, the indifference curve connecting these compound lotteries must be parallel to that connecting the simple lotteries L and L' .

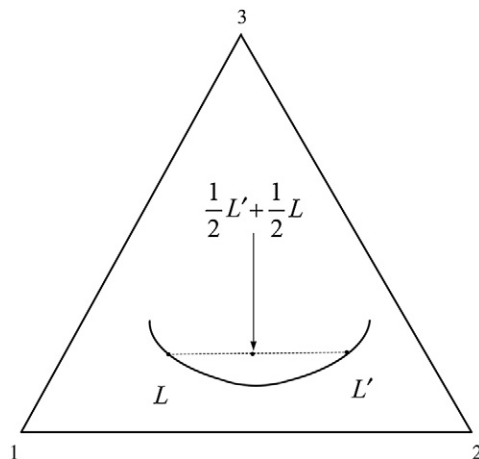


Figure 5.12
Curvy indifference curves violate the IA

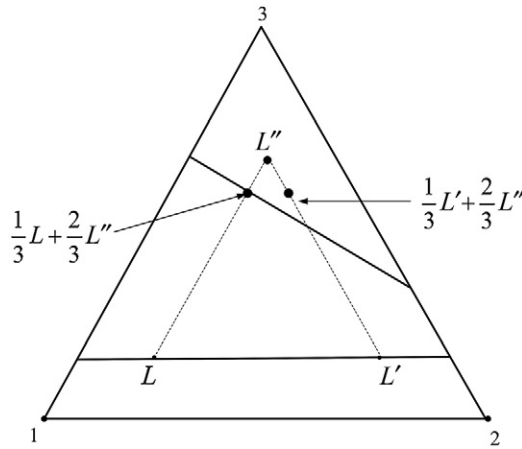


Figure 5.13
Nonparallel indifference curves violate the IA.

5.3 Violations of the IA

Despite the intuitive appeal of the independence axiom, individuals confronting choices over lotteries in experimental settings seem to violate it, as we show in the next examples.

Allais's paradox Consider a lottery over three possible monetary outcomes: a first prize of US\$2.5 million, a second prize of half a million dollars, and a third prize of zero dollars. The decision maker is initially asked to choose between lotteries L_1 and L'_1 , where

$$L_1 = (0, 1, 0) \quad \text{and} \quad L'_1 = \left(\frac{10}{100}, \frac{89}{100}, \frac{1}{100} \right).$$

Intuitively, lottery L_1 assigns full probability to the second prize (half a million dollars), whereas lottery L'_1 assigns a 89 percent probability to this prize, 10 percent probability to the first prize (2.5 million dollars), and 1 percent to the third prize (zero dollars). Which lottery would you prefer, L_1 or L'_1 ? (To participate, write your preference on a piece of paper before continuing.) The decision maker is then asked to select one lottery among the following two:

$$L_2 = \left(0, \frac{11}{100}, \frac{89}{100} \right) \quad \text{and} \quad L'_2 = \left(\frac{10}{100}, 0, \frac{90}{100} \right),$$

where L_2 assigns a 11 percent probability to the half million prize while lottery L'_2 assigns a 10 percent to the 2.5 million prize. Which lottery would you choose, L_2 or L'_2 ? (Practice also writing this down.) Let us now analyze which are the most common selections in controlled experiments. Interestingly, more than 50 percent of the students confronted with these two choices preferred lottery L_1 to L'_1 , but L'_2 to L_2 . (This result has been repeatedly observed in various countries, and among subjects with different educational backgrounds.) Let us next show why this preference relation violates the IA. If the decision maker's preferences over lotteries satisfy all previous assumptions (and hence can be captured by an expected utility function), the fact that L_1 is strictly preferred to L'_1 implies that

$$u_{0.5} > \frac{10}{100}u_{2.5} + \frac{89}{100}u_{0.5} + \frac{1}{100}u_0,$$

where $u_{0.5}$, $u_{2.5}$, and u_0 denotes the utility that the decision maker obtains from outcome 0.5 million, 2.5 million, and zero dollars, respectively. By the IA, we can add $\frac{89}{100}u_0 - \frac{89}{100}u_{0.5}$ to both sides and obtain

$$u_{0.5} + \left(\frac{89}{100}u_0 - \frac{89}{100}u_{0.5} \right) > \frac{10}{100}u_{2.5} + \frac{89}{100}u_{0.5} + \frac{1}{100}u_0 + \left(\frac{89}{100}u_0 - \frac{89}{100}u_{0.5} \right),$$

which we simplify as

$$\frac{11}{100}u_{0.5} + \frac{89}{100}u_0 > \frac{10}{100}u_{2.5} + \frac{90}{100}u_0.$$

Since the left-hand side coincides with the expected utility of lottery L_2 and the right-hand side represents the expected utility of lottery L'_2 , we can infer that $L_2 \succ L'_2$. Hence, if the decision maker prefers L_1 to L'_1 , the IA implies that he must prefer L_2 to L'_2 . (Did your own choices violate the IA?) The dissonance between theoretical predictions (as described by the IA) and people's actual choices over lotteries has produced several reactions from the literature:

1. *Approximation to rationality* One reaction to the Allais paradox considers that people might violate the IA the first time (or the first few times) they are confronted with choices among different lotteries. However, they are capable of adapting, and we should not expect subjects to continue violating the IA after a short period of time.

2. *Little economic significance* Another reaction to the Allais paradox says that the lotteries presented to subjects involve probabilities that are close to zero and one, something that rarely occurs in real economic settings.
3. *Regret theory* Some subjects justify their choice of lottery L_1 over L'_1 claiming they did not “want to regret a sure win of half a million dollars!” Such justification has led to the development of regret theory regarding choices under uncertainty. (See Exercise 5 for more details.)

Machina’s paradox Yet another setting where subjects in economic experiments have repeatedly violated the IA is in their preferences over certain outcomes.

For example, Enric prefers winning a trip to Barcelona (Spain) to winning a ticket to a movie about Barcelona, such as Woody Allen’s *Vicky Cristina Barcelona*; but he does prefer seeing the movie to staying at home. Let us now consider the following two lotteries over the three outcomes:

$$L_1 = \left(\frac{99}{100}, \frac{1}{100}, 0 \right) \quad \text{and} \quad L_2 = \left(\frac{99}{100}, 0, \frac{1}{100} \right).$$

Intuitively, the first lottery involves a 99 percent probability of winning a trip to Barcelona and a 1 percent probability of winning a ticket to the movie featuring Barcelona. The second lottery still maintains the same 99 percent probability of winning a trip to Barcelona but shifts the 1 percent probability toward the outcome in which the individual stays at home. One interesting characteristic of the IA is that we can infer a decision maker’s preference over two lotteries from his previous preferences over certain outcomes. Let us denote by T the trip to Barcelona, M the movie on Barcelona, and H staying home. Using the fact that this decision maker prefers T to M , which we express as $T \succ M$, we can construct the compound lotteries $\frac{99}{100}T + \frac{1}{100}M$ and $\frac{99}{100}M + \frac{1}{100}M$. From the IA we have that $\frac{99}{100}T + \frac{1}{100}M \succ \frac{99}{100}M + \frac{1}{100}M$.

Second, from the assumption that Enric prefers the movie rather than staying at home without watching the movie, i.e., $M \succ H$, we have

$$\frac{99}{100}M + \frac{1}{100}M \succ \frac{99}{100}M + \frac{1}{100}H$$

Hence, by transitivity, we obtain that $\frac{99}{100}T + \frac{1}{100}M \succ \frac{99}{100}M + \frac{1}{100}H$. Since the left-hand side coincides with lottery L_1 and the right-hand side coincides with L_2 , we can conclude that $L_1 \succ L_2$.

Hence a decision maker whose preference over lotteries satisfies the IA should prefer the first to the second lottery. By contrast, many subjects in experimental settings

prefer L_2 to L_1 , violating as a consequence the IA. As in the Allais paradox, many subjects explain choosing L_2 over L_1 because of the disappointment they would experience in the case of losing the trip (to Barcelona, Venice, or any other tourist destination) and having to watch a movie about the city instead.

In both lottery examples, the subjects' behaviors are inconsistent with the IA. Can we still rely on the IA as a sensible assumption about individuals' preferences among lotteries? One way to answer this question is instead by asking ourselves: What would happen to individuals whose behavior systematically violates the IA? As we describe next, such types of individuals would be weeded out of the market because they would be open to accepting so-called Dutch books, which would propel them to a sure loss of money.

Example 5.5: Dutch books Say an individual prefers lottery L to L' and lottery L to L'' , that is, $L \succ L'$ and $L \succ L''$. If his preferences satisfy the IA, he should prefer the compound lottery $\alpha L + (1 - \alpha)L$ for every $\alpha \in (0, 1)$, which coincides with the simple lottery L , rather than the compound lottery $\alpha L + (1 - \alpha)L'$. However, in violation of the IA, he does prefer $\alpha L + (1 - \alpha)L' \succ L$.

Hence, if we present the individual with the chance to trade lottery L for compound lottery $\alpha L + (1 - \alpha)L'$, which produces lottery L with probability α and lottery L' with probability $(1 - \alpha)$, for a small fee (x dollars) he would accept such a trade. As soon as the first stage of the compound lottery is over, the individual will have either lottery L or L' (i.e., the realization of the compound lottery gives him either L or L'). If the realization of the lottery is L' , we can again offer him to pay $\$y$ in order to trade L' for L , which he accepts since $L \succ L'$. If the realization is instead L , then we can offer him the compound lottery $\alpha L + (1 - \alpha)L'$ for a similar fee $\$y$, which he will accept since $\alpha L + (1 - \alpha)L' \succ L$. Thus, after both trades, he has paid two fees ($x + y$ dollars) and ended up exactly where he began. We could start the cycle again, extracting more money from this individual. Hence individuals who systematically violate the IA in their choices among risky lotteries can be exploited by heartless microeconomists, and they are ultimately weeded out of the marketplace. For this reason we assume that preferences satisfy the IA in most of our economic applications. ■

Further reading If you are interested in more violations of the axioms in expected utility theory and the theories that have been proposed to account for such observed behavior, you will enjoy the next section in this chapter, where we briefly describe the most salient of these theories. For more details, see the review article by Chris Starmer

(2000) and the book *Choices, Values and Frames* (2000) by Nobel prize winners Daniel Kahneman and Amos Tversky, Cambridge University Press. For a more detailed examination of the theoretical axioms in choice under uncertainty, see Itzhak Gilboa's (2009) textbook.

5.4 Behavioral Theories That Modify Expected Utility Theory

Several theories have been advanced to account for the experimental anomalies that standard expected utility theory cannot explain, as those in the Allais paradox. We briefly list some of them below.

5.4.1 Weighted Utility

Initially developed by Chew and McCrimmon (1979) and extended by Fishburn (1983), this theory assigns a weight function $g(\cdot)$ to each of the payoffs that the individual can obtain in the lottery:

$$V(L) = \sum_{x \in C} w_i \cdot u(x_i),$$

where

$$w_i = \frac{g(x_i)p(x_i)}{\sum_{x \in C} g(x_i)p(x_i)} \quad \text{and} \quad g: C \rightarrow \mathbb{R}.$$

In words, the utility of outcome $x_i \in C$, $u_i(x_i)$, is weighted not only according to its probability $p(x_i)$ but also according to the outcome x_i itself through function $g: C \rightarrow \mathbb{R}$. For instance, in a gamble between two payoffs x_1 and x_2 with probabilities p and $1-p$, respectively, the weighted utility is

$$\begin{aligned} V(L) &= w_1 u(x_1) + w_2 u(x_2) \\ &= \frac{g(x_1)p}{g(x_1)p + g(x_2)(1-p)} u(x_1) + \frac{g(x_2)(1-p)}{g(x_1)p + g(x_2)(1-p)} u(x_2). \end{aligned}$$

Now, if $g(x_i) = g(x_j)$ for any two payoffs $x_i \neq x_j$, this function collapses to a standard expected utility function $V(L) = pu(x_1) + (1-p)u(x_2)$. The weighted utility function relies on the same axioms as expected utility, except for the IA, which is relaxed to the following “weak independence axiom” due to Camerer (1989):

Weak independent axiom If an individual is indifferent between lotteries L_1 and L_2 , we can find a pair of probabilities α and α' such that he is indifferent between the compound lotteries

$$\alpha L_1 + (1 - \alpha)L_3 \quad \text{and} \quad \alpha' L_2 + (1 - \alpha')L_3$$

for every probability $\alpha \in (0, 1)$ and for every lottery L_3 .

The standard IA becomes a special case of this axiom if probabilities α and α' coincide; but otherwise, this weakened version of the axiom is more general.

5.4.2 Rank-Dependent Utility

An alternative theory that follows a similar approach to that of weighted utility theory by distorting probabilities so that they enter nonlinearly into the value function of the individual.

However, the theory differs in two dimensions: first, it requires that we rank the outcomes $x_1, x_2, \dots, x_n$ from worst (x_1) to best (x_n) and, second, that we apply a probability weighting function w_i to every outcome i , where

$$w_i = \pi(p_i + \dots + p_n) - \pi(p_{i+1} + \dots + p_n) \\ \text{and } w_n = \pi(p_n) .$$

Here $\pi(\cdot)$ is a nondecreasing transformation of cumulative probabilities with $\pi(0)=0$, and $\pi(1)=1$. In addition, $\pi(p_i + \dots + p_n)$ is the subjective weight that the individual assigns to receiving a payoff at least as good as x_i , whereas $\pi(p_{i+1} + \dots + p_n)$ is the subjective weight that he assigns to receiving a payoff strictly better than x_i . Hence w_i represents the subjective probability weight on receiving a payoff of exactly x_i . We just need to multiply the utility from each outcome x_i times its associated weighted probability w_i , yielding a rank-dependent utility

$$V(L) = \sum_{x \in C} w_i \cdot u(x_i) .$$

In the special case of a simple gamble between two outcomes, x_1 and x_2 , where $x_2 > x_1$, the rank-dependent utility becomes

$$V(L) = w(p)u(x_1) + (1 - w(p))u(x_2) ,$$

where p denotes the probability of outcome x_1 and $w(p)$ is the subjective probability weight that the individual assigns to this outcome.

Intuitively, this model allows individual decision makers to attach a different weight to each outcome, depending on how good or bad it is relative to other outcomes—as

opposed to expected utility theory models in which individuals attach the same weight to all outcomes. As suggested by Quiggin (1993), a pessimistic individual can use a concave transformation function $\pi(\cdot)$, yielding $\pi(p) \leq p$ for all p , thus assigning a higher (lower) probability weight on low (high) payoffs than p does. The opposite argument applies for an optimistic individual using a convex transformation function $\pi(\cdot)$, which implies that $\pi(p) \geq p$. Figure 5.14 depicts the transformation function for pessimistic and optimistic individuals.

Empirical evidence nevertheless does show that individuals are not pessimistic or optimistic for all values of p (as both panels in figure 5.14 would suggest); they instead exhibit an S-shaped transformation function as the one in figure 5.15 (see Prelec 1998 for a survey of experimental results). For illustration purposes, the figure depicts $\pi(p) = e^{(-\ln(p))^\alpha}$, evaluated at $\alpha = 1/2$, but generally α can satisfy $\alpha \in (0, 1)$. Intuitively, individuals may be pessimistic in rare outcomes (those with a low probability, i.e., $p < \bar{p}$ in the figure) for which they do not have much previous experience, but be optimistic for outcomes they frequently encounter. Interestingly, rank-dependent utility theory relies on the same axioms as expected utility theory, except for the IA which is replaced by co-monotonic independence (for more details on these axioms, see the review by Machina and Siniscalci 2014).

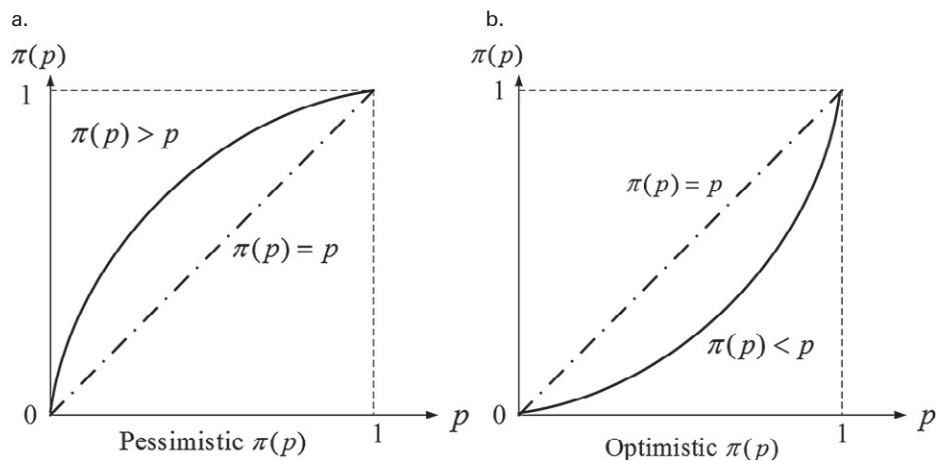


Figure 5.14
Transformation function $\pi(p)$ in rank-dependent utility

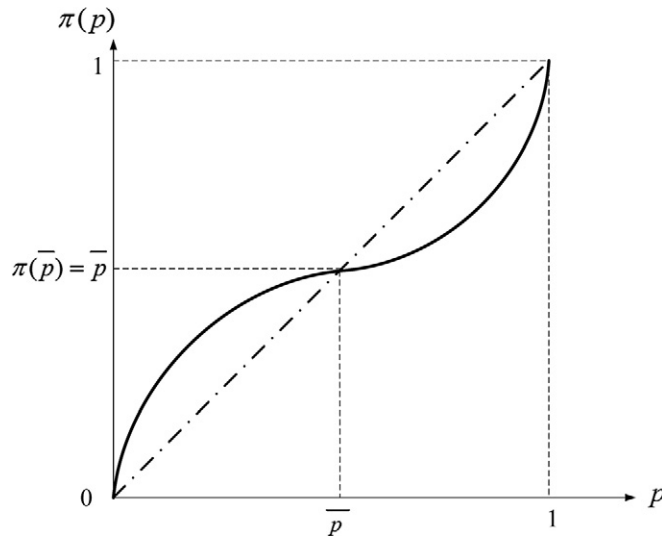


Figure 5.15
S-shaped transformation function $\pi(p)$ in rank-dependent utility

5.5 Money Lotteries

In the following sections we restrict our attention to lotteries over monetary outcomes. Now the set of possible outcomes, C , satisfies $C = \mathbb{R}$; hence the outcomes are a continuous variable, $x \in \mathbb{R}$. This allows us to describe money lotteries as the following cumulative distribution function (cdf):

$$F(x) = \text{prob}\{y \leq x\} \quad \text{for all } y \in \mathbb{R}.$$

In words, $F(x)$ represents the probability that the realized payoff y is less than or equal to x . Figure 5.16a illustrates an example of a money lottery that assigns the same probability weight to every possible payoff, which can therefore be represented with a uniform cdf $F(x) = x$. Figure 5.16b depicts a money lottery that assigns a smaller probability weight to the initial monetary values (approximately before \$0.5) than to the last values (beyond \$0.5).

The preceding examples consider continuous probability distributions. The decision maker can, of course, face a money lottery that behaves according to a discrete probability distribution, as figure 5.17 illustrates. We express this as

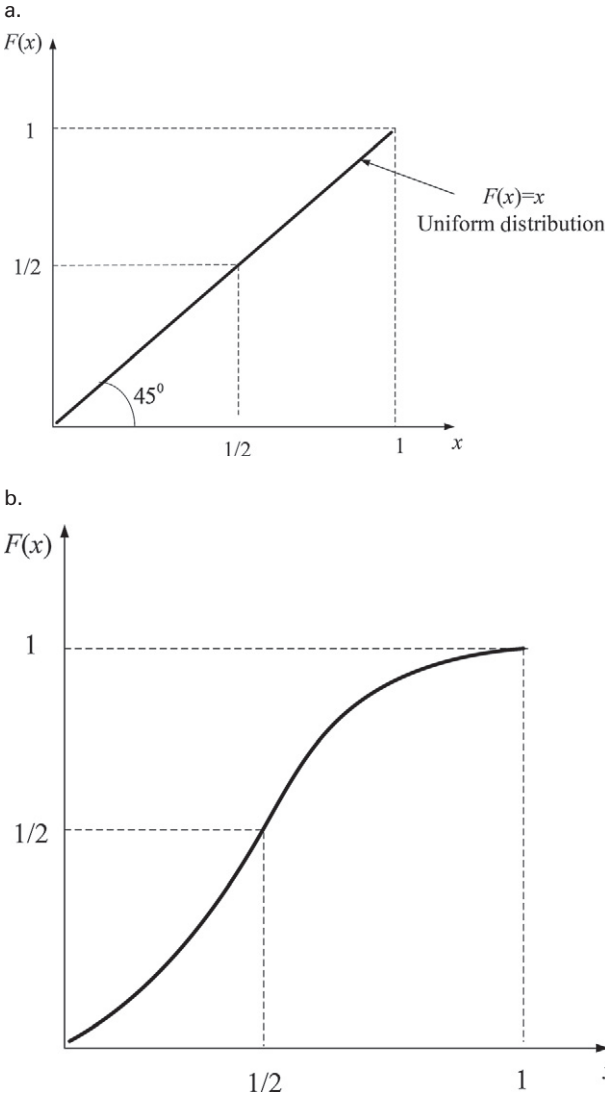


Figure 5.16
(a) Money lottery with a uniform distribution $F(x) = x$. (b) Nonuniform money lottery

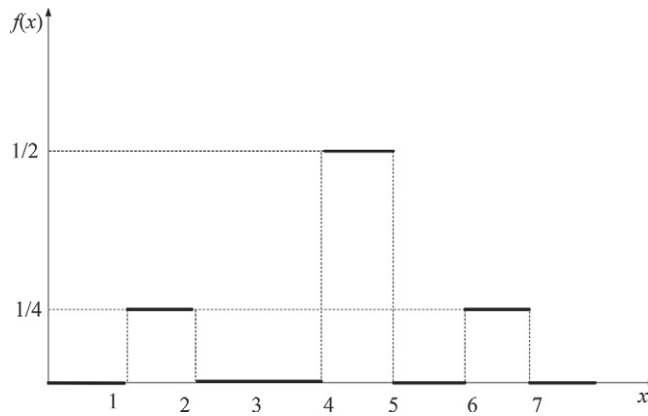


Figure 5.17
Example of a discrete distribution function

$$F(x) = \begin{cases} 0 & \text{if } x < 1, \\ \frac{1}{4} & \text{if } x \in [1, 4), \\ \frac{3}{4} & \text{if } x \in [4, 6), \\ 1 & \text{if } x \geq 6. \end{cases}$$

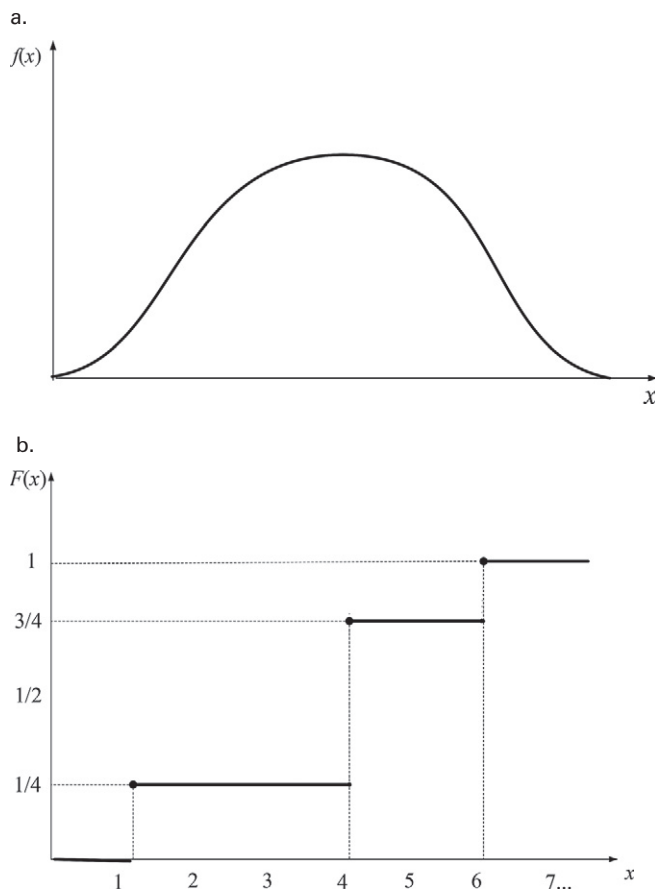
If in addition there is a density function $f(x)$ associated with the cdf $F(x)$, we can then represent the money lottery as

$$F(x) = \int_{-\infty}^x f(t) dt.$$

Figure 5.18a illustrates the density function $f(x)$ for a continuous cdf, which provides a clear representation of the probability weight on different monetary amounts. For this example, intermediate payoffs are likely. Figure 5.18b depicts the density function of the discrete cdf $F(x)$ in figure 5.18b, where the most probability weight is concentrated in the interval of \$4 to \$5.

Thus far we have only discussed simple monetary lotteries, but we can also represent compound lotteries. For example, if the list of cdf's $F_1(x)$, $F_2(x)$, ..., $F_K(x)$ represents K simple money lotteries, each occurring with probability $\alpha_1, \alpha_2, \dots, \alpha_K$, respectively, then the compound lottery can be represented as the following cdf:

$$F(x) = \sum_{k=1}^K \alpha_k F_k(x),$$

**Figure 5.18**

(a) Density function of the continuous cdf. (b) Density function of a discrete cdf

which intuitively represents the expected (monetary) value of playing the K simple money lotteries. For simplicity, we now consider money lotteries to be distributed over nonnegative monetary amounts $C \in \mathbb{R}_+$.⁸ We can express the expected utility that the decision maker obtains from playing a money lottery $F(x)$, with associated density function $f(x)$, as follows:

$$U(F) = \int u(x) f(x) dx, \quad \text{or} \quad \int u(x) dF(x),$$

⁸ This just implies a normalization that shifts all possible payoffs in the lottery, such as summing a constant P to all of them, where P represents the smallest negative payoff that the decision maker can obtain in the lottery. This normalization guarantees that all resulting payoffs are zero or positive.

where $u(x)$ denotes the utility that the decision maker obtains when the lottery gives him a monetary amount of x dollars.⁹ Note that $U(F)$ is the mathematical expectation of the values of $u(x)$, over all possible values of x . Furthermore this expression is linear in the probabilities. Indeed, in the case that the cdf is a discrete probability distribution (as in the previous examples), we can find the expected utility from playing such a money lottery by writing $p_1u(x_1) + p_2u(x_2) + \dots$. Importantly, this expected utility representation is sensitive not only to the mean of the distribution but also to the variance and higher moments of the distribution of monetary payoffs. We show this property of the expected utility function in the next example.¹⁰

Example 5.6: Expected utility that only depends on two moments Consider a decision maker with a Bernoulli utility function $u(x) = \beta x^2 + \gamma x$. We will show that his expected utility of any given lottery $F(x)$ is determined by the mean and variance of the money lottery alone:

$$\begin{aligned} EU &= \int u(x) dF(x) = \int [\beta x^2 + \gamma x] dF(x) \\ &= \beta \int x^2 dF(x) + \gamma \int x dF(x) = \beta E(x^2) + \gamma E(x). \end{aligned}$$

By definition, we know that the variance of a random variable x is expressed as $\text{Var}(x) = E(x^2) - E(x)^2$. Hence $E(x^2) = \text{Var}(x) + E(x)^2$.

Substituting $E(x^2)$ in the expression above, we obtain the expected utility of the lottery as

$$EU = \underbrace{\beta \text{Var}(x) + \beta E(x)^2}_{\beta E(x^2)} + \gamma E(x).$$

As a consequence the expected utility is determined by the mean, $E(x)$, and variance, $\text{Var}(x)$, alone. ■

Thus far we have imposed a relatively limited set of assumptions on the decision maker's Bernoulli utility function, $u(x)$: the utility he obtains from certain monetary

⁹ If there is a density function $f(x)$ associated to the cdf $F(x)$, we can use either of the expressions above. Otherwise, we can only use the latter. In addition, note that we did not write the intervals of integration. We hereafter assume that the integral is defined over the full range of possible realizations of x , namely from zero to infinity.

¹⁰ For more examples on expected utility maximization problems, see, for instance, exercises 8, 9, 11 and 22–25 at the end of this chapter.

outcomes is increasing in money and is continuous. We need to also impose that $u(x)$ is bounded. Otherwise, we may end up in relatively absurd situations, such as that illustrated in the so-called St. Petersburg–Menger paradox, which we present next.

St. Petersburg–Menger paradox Using an unbounded Bernoulli utility function, $u(x)$, we can find an amount of money x_m such that $u(x_m) > 2^m$ for any integer m .

To do so, we consider a lottery in which we toss a coin repeatedly until tails comes up. We then give a monetary amount x_m if tails comes up at the m th toss. Since the probability that tails comes up in the m th toss is

$$\frac{1}{2} \frac{1}{2} \frac{1}{2} \dots (m \text{ times}) = \frac{1}{2^m},$$

the expected utility from playing such a lottery is

$$EU = \sum_{m=1}^{\infty} \frac{1}{2^m} u(x_m).$$

But because $u(x_m) > 2^m$ by assumption, we then have that

$$EU = \sum_{m=1}^{\infty} \frac{1}{2^m} u(x_m) \geq \sum_{m=1}^{\infty} \frac{1}{2^m} 2^m = \sum_{m=1}^{\infty} 1 = +\infty,$$

where the expression on the right is infinitely large. Hence this individual would be willing to pay an infinite amount of money to be able to play this lottery, since the expected utility he obtains from playing the lottery satisfies $EU \geq +\infty$. Therefore, to avoid these cases, we assume that the Bernoulli utility function, $u(x)$, must be bounded.¹¹

5.6 Measuring Risk Preferences

In this section we evaluate individual preferences toward risky lotteries. We start with the measure of risk aversion.

¹¹ Alternatively, we can avoid situations such as that described in the St. Petersburg–Menger paradox by checking that the distribution function (rather than the Bernoulli utility function) does not allow for this type of paradoxes. (For more about this approach, see Nicholson and Snyder 2011), 203–205.)

5.6.1 Risk Aversion

Risk aversion We say that an individual's utility exhibits risk aversion if, for any money lottery $F(\cdot)$,

$$\int u(x) dF(x) \leq u\left(\int x dF(x)\right).$$

If this relationship holds with equality, we denote this individual as risk neutral. If, instead, the sign of the inequality is reversed ($\geq$), we denote him as a risk lover.

Intuitively, the utility that this individual obtains from receiving the *expected monetary value* of playing lottery $F(\cdot)$, at the right-hand side of the inequality above, is higher than the *expected utility* from playing such a lottery, as captured by the left-hand side, that is, $u(EV) \geq EU$. Figure 5.19 illustrates this case. In particular, it considers a lottery with two possible outcomes: \$1 and \$3 that are equally likely. In the figure we depict the utility from outcomes \$1 and \$3, $u(1)$ and $u(3)$, respectively, by mapping the monetary amounts \$1 and \$3 into their utility values $u(1)$ and $u(3)$, on the vertical axis. We then find the *expected value* of the lottery (\$2, i.e., $\frac{1}{2}\$1 + \frac{1}{2}\3) and map it into the utility function, obtaining $u(2)$. This is what we refer to as the utility from the expected monetary value of the lottery, represented by $u(EV)$.

We can now analyze the expected utility from playing the lottery, which provides the individual with a utility of $u(1)$, with probability $\frac{1}{2}$, or a utility of $u(3)$, with a

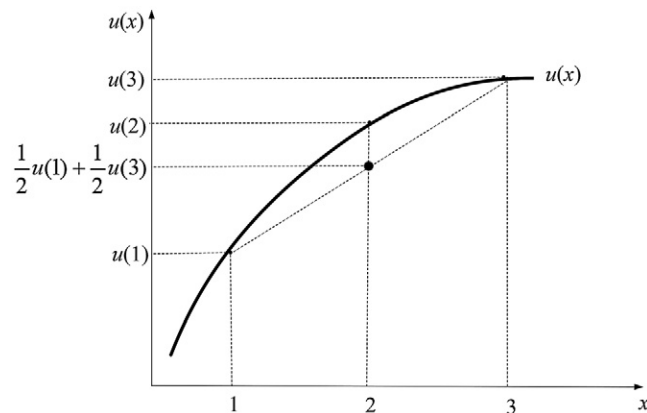


Figure 5.19
Risk-averse individual

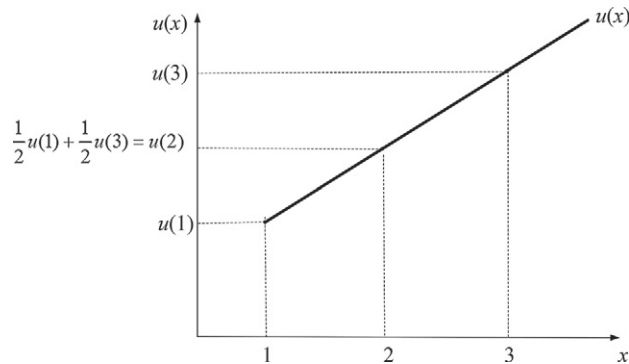


Figure 5.20
Risk-neutral individual

probability of $\frac{1}{2}$. Connecting $u(1)$ and $u(3)$, and examining the midpoint of this line, $\frac{1}{2}u(1) + \frac{1}{2}u(3)$, we obtain the *expected utility* of playing the lottery. In this example, the individual's utility function is concave in x , such as $u(x) = x^\alpha$, where $0 < \alpha < 1$, implying that the utility from the expected value of the lottery, $u(2)$, is higher than the expected utility from playing the lottery, $\frac{1}{2}u(1) + \frac{1}{2}u(3)$. We can therefore conclude that this individual's utility exhibits risk aversion.¹² For instance, if $u(x) = \sqrt{x}$, then $EU = \frac{1}{2}\sqrt{1} + \frac{1}{2}\sqrt{3} = 1.36$; which is lower than $u(EV) = u(\frac{1}{2} \times 1 + \frac{1}{2} \times 3) = u(2) = \sqrt{2} = 1.41$.

Figure 5.20 depicts a risk neutral individual, as is evident by the fact that $u(x)$ is linear in x . In this case the utility from the expected value of the lottery, $u(2)$, coincides with the expected utility of playing the lottery, $\frac{1}{2}u(1) + \frac{1}{2}u(3)$. In contrast, if an individual is risk lover, as figure 5.21 illustrates, the utility from the expected value of the lottery, $u(2)$, is lower than the expected utility from playing the lottery, $\frac{1}{2}u(1) + \frac{1}{2}u(3)$. Here we see that $u(x)$ is convex in x , in that $u(x) = x^\alpha$, where $\alpha > 1$.

An alternative way to measure risk aversion is by finding the "certainty equivalent" of a lottery, as we describe next.

¹² This definition of risk aversion is in fact a direct application of Jensen's inequality, which suggests a strong connection between the concavity of an individual's utility function and his degree of risk aversion. We return to this connection when we discuss the Arrow–Pratt's coefficient of absolute risk aversion.

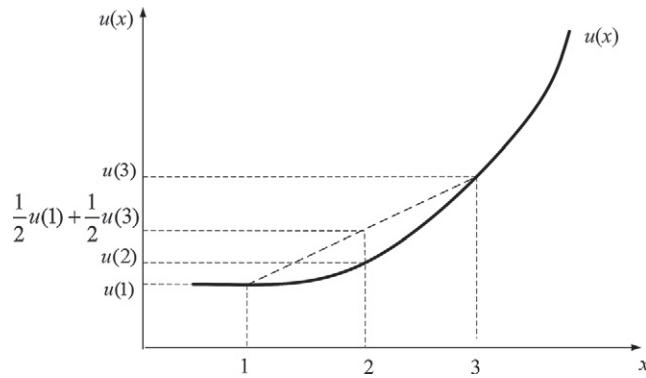


Figure 5.21
Risk-loving individual

5.6.2 Certainty Equivalent

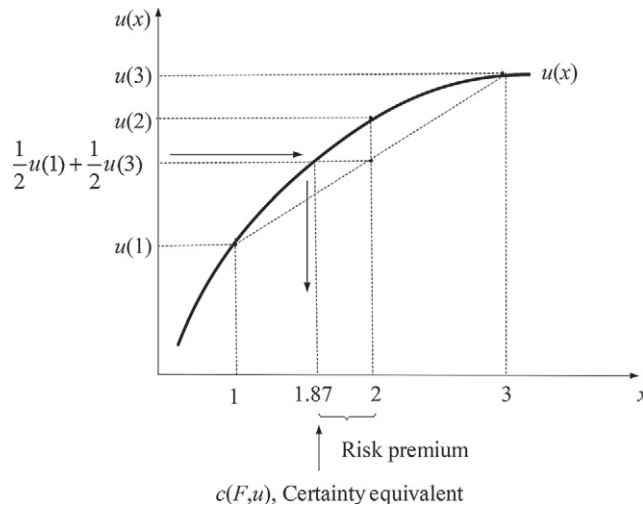
Certainty equivalent The certainty equivalent of money lottery $F(\cdot)$ for an individual with utility function $u(\cdot)$, $c(F, u)$, is the amount of money for which the individual is indifferent between playing lottery $F(\cdot)$ and accepting a certain (sure) amount of money $c(F, u)$.

More compactly, the certainty equivalent can be expressed as the dollar amount $c(F, u)$ that satisfies

$$u(c(F, u)) = \int u(x) dF(x),$$

where the left-hand side measures the utility from receiving the monetary amount $c(F, u)$ with certainty, while the right-hand side denotes the expected utility that this individual obtains from playing lottery $F(\cdot)$. Figure 5.22 illustrates the certainty equivalent for a risk-averse individual. Specifically, $c(F, u)$ is the amount of money that helps the individual to reach the same utility as if he played lottery $F(\cdot)$, given that his utility function is $u(\cdot)$. Because he is risk averse, the certainty equivalent $c(F, u)$ is below the expected value of the lottery, \$2. Graphically, $c(F, u)$ is found by moving from the expected utility of the lottery in the vertical axis to the monetary amount that would yield this utility level in the horizontal axis (represented with arrows in Figure 5.22). In particular, for this numerical example, $c(F, u)$ can be found by solving $u(c(F, u)) = \frac{1}{2}u(1) + \frac{1}{2}u(3)$.¹³

¹³ If, for instance, $u(x) = \sqrt{x}$, then the certainty equivalent $c(F, u)$ solves $\sqrt{c(F, u)} = \frac{1}{2}\sqrt{1} + \frac{1}{2}\sqrt{3}$, or $\sqrt{c(F, u)} = 1.36$. Solving for $c(F, u)$, we obtain a certainty equivalent of $c(F, u) = 1.87$ (after rounding).

**Figure 5.22**

Certainty equivalent and risk premium for a risk-averse individual

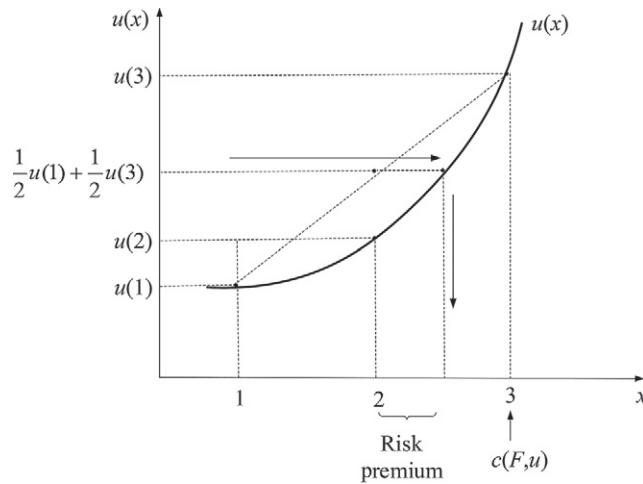
The difference between the expected value of the lottery and the certainty equivalent that a risk-averse individual would be willing to accept in order to avoid the risky lottery is also used as a measure of how risk averse a particular individual is. Specifically, this measure is commonly referred as the “risk premium” of a lottery, RP , and is defined as

$$RP = EV - c(F, u).$$

Figure 5.22 also includes the risk premium that this individual is willing to bear in order to avoid the risky lottery; that is, he is willing to accept a certainty equivalent below the expected value of the lottery. In this example, he is willing to accept a certain monetary amount of only \$1.87 rather than playing the lottery, thus implying a risk premium of $\$2 - \$1.87 = \$0.13$.

In the case of a risk lover, the previous ranking is reversed, as figure 5.23 illustrates. Indeed the certainty equivalent $c(F, u)$ lies above the expected value of the lottery, \$2. As a consequence the risk premium for this individual, $RP = EV - c(F, u)$ is actually negative, since $EV < c(F, u)$. Intuitively, this implies that the individual would have to receive an amount of money *above* the expected value of the lottery in order to convince him to stop playing the lottery (he certainly loves risk!).

In the case of a risk-neutral decision maker, the certainty equivalent $c(F, u)$ coincides with the expected value of the lottery, \$2, and therefore the risk premium is zero, since $RP = EV - c(F, u)$. Clearly, this individual is not willing to accept money to avoid

**Figure 5.23**

Certainty equivalent and risk premium for a risk lover

playing the lottery (as for risk-averse individuals), nor must we compensate him in order to stop playing the lottery (as a risk lover). Figure 5.24 depicts this case.¹⁴

All the measures of riskiness discussed above evaluate risk preference using monetary payoffs. The next measure of risk aversion, in contrast, uses probabilities.

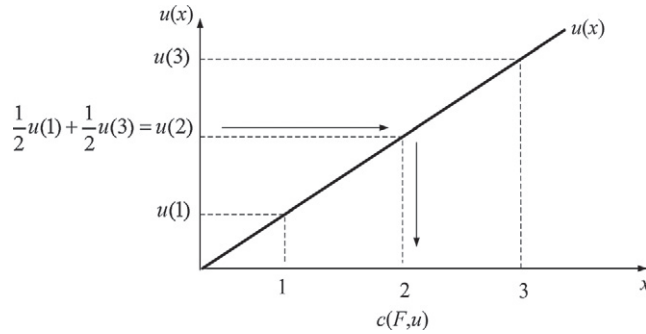
5.6.3 Probability Premium

Probability premium The probability premium measures the excess in winning probability over fair odds (equally likely outcomes) that makes the individual indifferent between the certainty outcome x and a gamble between the two outcomes $x + \varepsilon$ and $x - \varepsilon$. That is,

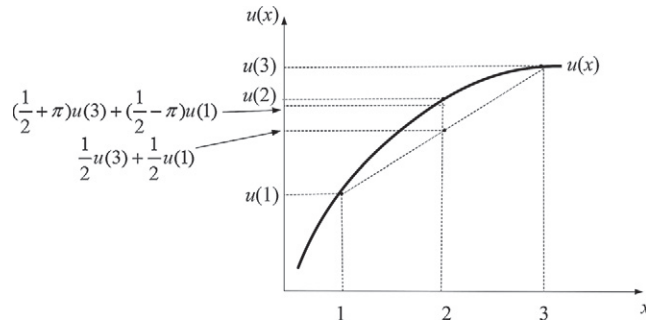
$$u(x) = \left[\frac{1}{2} + \pi(x, \varepsilon, u) \right] u(x + \varepsilon) + \left[\frac{1}{2} - \pi(x, \varepsilon, u) \right] u(x - \varepsilon).$$

Intuitively, this implies that a risk-averse individual, in order to be attracted to play a particular lottery, must be given *better* than fair odds, since otherwise he would not accept the risk associated with the lottery. This intuition is graphically represented in figure 5.25 for our ongoing example of the lottery between monetary amounts \$1 and \$3. First,

¹⁴ For an introductory parametric example on how to find the expected utility, certainty equivalent, and risk premium of a lottery, see exercise 7 at the end of the chapter.

**Figure 5.24**

Certainty equivalent and risk premium for a risk-neutral individual

**Figure 5.25**

Probability premium

we map \$1 and \$3 into the utility function, obtaining $u(1)$ and $u(3)$. We find the expected value of the lottery, \$2, and map it into the utility function, yielding $u(2)$. Then we find the “extra winning probability” π (extra probability of outcome \$3 occurring) that the risk-averse decision maker needs in order to make the EU of the lottery raise until it coincides with the utility from the expected value of the lottery, $u(2)$, that is

$$u(2) = \left(\frac{1}{2} + \pi\right) \cdot u(3) + \left(\frac{1}{2} - \pi\right) \cdot u(1).$$

Graphically, note that the EU from the lottery with fair odds (equally likely outcomes) $\frac{1}{2}u(1) + \frac{1}{2}u(3)$ lies below the lottery in which the winning probability (the likelihood of outcome \$3 occurring) has been increased by π .¹⁵

¹⁵ The probability premium of a lottery is also referred as the “insurance premium.”

Given these different measures of risk aversion, we next establish a connection between them. In particular, the following properties are equivalent:¹⁶

1. The decision maker is risk averse.
2. The utility function is concave, $u''(x) \leq 0$.
3. The certainty equivalent is lower than the expected value of the lottery, written as $c(F, u) \leq EV$.
4. The risk premium is positive, written as $RP = EV - c(F, u) > 0$, which implies that $EV \geq c(F, u)$.
5. The probability premium is positive for all x and ε , $\pi(x, \varepsilon, u) \geq 0$.

5.7 Arrow–Pratt Coefficients of Absolute and Relative Risk Aversion

5.7.1 Absolute Risk Aversion

In this section we examine other forms of measuring risk aversion. In particular, we focus on the connection between risk aversion and the concavity of a decision maker's utility function. We start by examining the Arrow–Pratt coefficient of absolute risk aversion, $r_A(x)$:

$$r_A(x) = -\frac{u''(x)}{u'(x)}.$$

The equation tells us that, as the curvature of the utility function gets greater, captured by $u''(x)$, the coefficient $r_A(x)$ gets larger.¹⁷ However, we cannot simply use $u''(x)$ to measure an individual's risk aversion. In particular, such a measure is not invariant to positive linear transformations of the utility function. For instance, if we apply a linear transformation $v(x) = \beta u(x)$, then the concavity of the transformed function, $v''(x) = \beta u''(x)$, is affected by the linear transformation, whereas coefficient $r_A(x)$ is invariant, since

$$r_A(x) = -\frac{\beta u''(x)}{\beta u'(x)} = -\frac{u''(x)}{u'(x)}.$$

Example 5.7: CARA utility function Consider utility function $u(x) = -e^{-ax}$ where $a > 0$. The Arrow–Pratt coefficient of absolute risk aversion, $r_A(x)$, is

¹⁶ Exercise 21 at the end of the chapter asks you to prove points 2 and 3 of the following list.

¹⁷ Actually $r_A(x)$ is higher for individuals with more concave utility functions. For a proof of this result, see exercise 16 at the end of the chapter.

$$r_A(x) = -\frac{u''(x)}{u'(x)} = -\frac{-a^2 e^{-ax}}{a e^{-ax}} = a \quad \text{for all } x,$$

where $r_A(x)$ is constant in the individual's wealth level, x . As a consequence, the literature refers to this utility function, $u(x) = -e^{-ax}$, as the constant absolute risk aversion (CARA) utility function. ■

Coefficient $r_A(x)$ can, however, change in wealth. In particular, if $r_A(x)$ decreases as we increase wealth x , we say that such utility function satisfies decreasing absolute risk aversion (DARA), which we write as $\partial r_A(x)/\partial x < 0$. Intuitively, this implies that wealthier individuals are willing to bear more risk than poorer individuals.

Example 5.8: DARA utility function Some economic applications consider DARA utility functions where $r'_A(x) \leq 0$. In particular, this derivative is

$$r'_A(x) = -\frac{(u'''u' - u''u'')}{(u')^2} = -\frac{u'''}{u'} + \left(\frac{u''}{u'}\right)^2.$$

For $r'_A(x)$ to be negative, we need

$$0 \leq \left(\frac{u''}{u'}\right)^2 \leq \frac{u'''}{u'}.$$

Last, since $u' > 0$ by assumption, the inequality above simplifies to

$$0 \leq (u'')^2 \leq u''',$$

which ultimately implies that $u''' \geq 0$ must hold for utility function $u(x)$ to exhibit DARA. Typical examples of DARA utility functions are $u_1(x) = (x + \alpha)^\beta$, where $\alpha \geq 0$ and $\beta \in (0, 1)$, and $u_2(x) = \ln(x + \alpha)$, where $\alpha \geq 0$. For $u_1(x)$, we obtain

$$r_A(x) = -\frac{u''}{u'} = -\frac{(\beta-1)\beta(x+\alpha)^{\beta-2}}{\beta(x+\alpha)^{\beta-1}} = \frac{1-\beta}{x+\alpha},$$

which decreases in wealth, x . Similarly, for $u_2(x)$, we find

$$r_A(x) = -\frac{u''}{u'} = -\frac{-1/(x+\alpha)^2}{1/(x+\alpha)} = \frac{1}{x+\alpha},$$

which is also decreasing in x . There is empirical support for DARA utility functions, as individuals seem to be less willing to pay for insurance as their wealth increases (i.e., as if they regarded insurance and wealth as substitutable). ■

Example 5.9: IARA utility functions From previous sections we know that quadratic utility functions like

$$u(x) = \alpha + \beta x - \frac{c}{2}x^2$$

have an expected utility that can be expressed as a function of the lottery mean and variance alone. At this point, we seek to show that this type of utility function does not exhibit DARA. First note that $u' = \beta - cx$, $u'' = -c$, and $u''' = 0$. Hence these utility functions do not exhibit DARA. Alternatively, we can find the $r_A(x)$ and check that it is increasing in wealth. Specifically,

$$r_A(x) = -\frac{u''}{u'} = -\frac{-c}{\beta - cx} = \frac{c}{\beta - cx},$$

which is clearly increasing in wealth, x , since $r'_A(x) = c^2/(\beta - cx)^2 > 0$ for all x . ■

5.7.2 Relative Risk Aversion

To avoid the possibility that our measure of risk aversion changes depending on the particular wealth level at which it is evaluated, we often use the following *relative* risk aversion coefficient, $r_R(x)$, which is unaffected by the precise value of x at which wealth is evaluated:

$$r_R(x) = -x \cdot \frac{u''(x)}{u'(x)}.$$

Then, since $r_A(x) = -u''(x)/u'(x)$, we can then express $r_R(x)$ as $r_R(x) = x \cdot r_A(x)$. Utility functions with a coefficient of relative risk aversion, $r_R(x)$, that is constant in wealth, $\partial r_R(x)/\partial x = 0$, are commonly referred as the constant relative risk aversion (CRRA) utility function, as the following example illustrates.

Example 5.10: CRRA utility function An example of a utility function with a constant $r_R(x)$ coefficient is $u(x) = x^b$, since

$$r_R(x) = -x \cdot \frac{b(b-1)x^{b-2}}{bx^{b-1}} = -x \cdot \frac{(b-1)x^{b-1}}{x \cdot x^{b-1}} = 1-b$$

where $r_R(x)$ is constant in wealth, x . ■

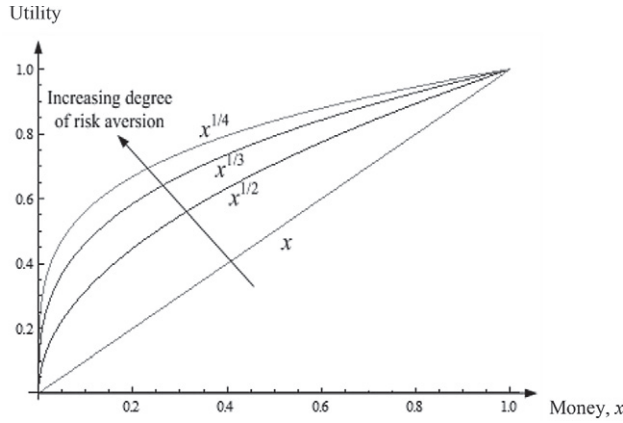


Figure 5.26
CRRA utility function

Hence differentiating $r_R(x)$ with respect to x , we obtain

$$\frac{\partial r_R(x)}{\partial x} = r_A(x) + x \cdot \frac{\partial r_A(x)}{\partial x}.$$

Therefore, if a utility function exhibits decreasing relative risk aversion, $\partial r_R(x)/\partial x < 0$, then it must also exhibit decreasing absolute risk aversion, $\partial r_A(x)/\partial x < 0$, while the converse is not necessarily true.¹⁸ For illustration purposes, figure 5.26 depicts a CRRA utility function $u(x) = x^b$ for $b = 1, 1/2, 1/3$, and $1/4$, yielding a coefficient of relative risk aversion, $r_R(x)$, of 1 (linear utility function) that increases, respectively, to $1/2, 2/3$, and $3/4$, making the utility function more concave.¹⁹

Rabin Paradox A common observation in controlled experiments is that the estimated Arrow–Pratt coefficient of relative risk aversion, $r_R(x)$, tends to be quite high for small risks. For instance, Holt and Laury (2002) report values of $r_R(x)$ close to 28,000 when individuals compare bets in which they can win small amounts of money, though other authors analyzing risk aversion for larger investments (e.g., stocks vs. bonds)

¹⁸ That is, since $\partial r_R(x)/\partial x = r_A(x) + x \cdot (\partial r_A(x)/\partial x)$, where $r_A(x) > 0$, if we observe that $\partial r_R(x)/\partial x < 0$, it must be because $\partial r_A(x)/\partial x$ is sufficiently negative to offset the positive term $r_A(x)$. However, if we only observe that $\partial r_A(x)/\partial x < 0$, we cannot guarantee that $\partial r_R(x)/\partial x$ is also negative.

¹⁹ Exercise 7 at the end of the chapter considers another family of utility functions that exhibit hyperbolic absolute risk aversion (HARA).

found significantly lower values of $r_R(x)$ (around 9). This enormous difference in individuals' coefficient of relative risk aversion, often referred to as the "Rabin paradox," cannot be explained using standard models of risk aversion as that in previous sections. Alternative theories of choice under uncertainty have been suggested to account for this anomaly, such as prospect theory (which we describe in section 5.9).²⁰

5.7.3 Strong Risk Aversion

Last we come to a more restrictive condition, often referred as "strong risk aversion." Introduced by Ross (1981), a utility function $u_A(\cdot)$ exhibits a stronger risk aversion than another utility function $u_B(\cdot)$ if there is a constant $\lambda > 0$, such that

$$\frac{u_A''(x_1)}{u_B''(x_1)} \geq \lambda \geq \frac{u_A'(x_2)}{u_B'(x_2)}.$$

If additionally $x_1 = x_2$, the condition above can be rewritten as

$$\frac{u_A''(x_1)}{u_A'(x_1)} \geq \frac{u_B''(x_1)}{u_B'(x_1)},$$

which implies that, if a function $u_A(\cdot)$ exhibits stronger risk aversion than $u_B(\cdot)$, then it also exhibits more absolute risk aversion than $u_B(\cdot)$.²¹

5.7.4 Relationships

We next establish an equivalence among the above measures of risk aversion. For two utility functions u_1 and u_2 , where u_2 is a concave transformation of u_1 (i.e., there exists an increasing concave function $\varphi(\cdot)$ such that $u_2(x) = \varphi(u_1(x))$ for all x , and as a consequence, u_2 is more concave than u_1), we have the following properties:

1. The coefficient of absolute risk aversion for $u_2(\cdot)$, increases as the utility function becomes more concave, in that $r_A(x, u_2) \geq r_A(x, u_1)$.

²⁰ Barberis et al. (2006), however, claim that prospect theory cannot fully explain such anomaly once we consider the time between the agent's decision whether to take a gamble and its realization, since the individual faces other risks during that time period.

²¹ The converse is not necessarily true. To see this, consider $u_i(x) = -e^{-a_i x}$, where $i = \{A, B\}$ and $a_A > a_B$. As described in example 5.7, $a_A > a_B$ implies that Arrow-Pratt coefficient of absolute risk aversion, $r_A(x)$, is larger for $u_A(\cdot)$ than $u_B(\cdot)$. However, $u_A'(x_2)/u_B'(x_2) = (a_A/a_B)e^{a_B - a_A x_2}$, while $u_A''(x_1)/u_B''(x_1) = (a_A/a_B)^2 e^{(a_B - a_A)x_1}$. Hence we can have $u_A''(x_1)/u_B''(x_1) < u_A'(x_2)/u_B'(x_2)$, thus violating strong absolute risk aversion. In particular, $u_A''(x_1)/u_B''(x_1) < u_A'(x_2)/u_B'(x_2)$ occurs when $a_A/a_B < e^{-(a_A - a_B)(x_2 - x_1)}$, which may hold if $x_2 < x_1$.

2. The certainty equivalent that a decision maker with utility function $u_2(\cdot)$ is willing to accept in order to avoid the lottery is lower than that of a decision maker with utility function $u_1(\cdot)$, that is, $c(F, u_2) \leq c(F, u_1)$ for every lottery $F(\cdot)$.
3. The probability premium that the individual with utility function $u_2(\cdot)$ needs to receive in order to accept playing lottery $F(\cdot)$ is higher than that of the individual with utility function $u_1(\cdot)$, in that $\pi(x, \varepsilon, u_2) \geq \pi(x, \varepsilon, u_1)$.
4. Whenever $u_2(\cdot)$ finds a lottery $F(\cdot)$ at least as good as a riskless outcome $\bar{x}$, then $u_1(\cdot)$ also finds such a lottery $F(\cdot)$ at least as good as $\bar{x}$. That is

$$EU_2 = \int u_2(x) dF(x) \geq u_2(\bar{x}) \quad \text{implies} \quad EU_1 = \int u_1(x) dF(x) \geq u_1(\bar{x}).$$

Figure 5.27 summarizes these results. Note in the figure that $u_1(\cdot)$ and $u_2(\cdot)$ are evaluated at the same wealth level x . The outcomes \$1 and \$3 are mapped into $u_1(\cdot)$ and into $u_2(\cdot)$ separately (see bullets in the figure). Connecting $u_1(1)$ and $u_1(3)$, we obtain the expected utility of playing the lottery for individual 1, and similarly for individual 2 where $EU_1 > EU_2$. Intuitively, the same lottery yields a larger expected utility for the individual with less risk-averse preferences, namely a less concave utility function. We can find the certainty equivalent for each individual, that is, the amount of money that provides each individual with the same utility as the expected utility he obtains from actually playing the lottery. As figure 5.27 shows, the certainty equivalent that individual 2 is willing to accept in order to avoid playing the lottery, $c(F, u_2)$ is lower than

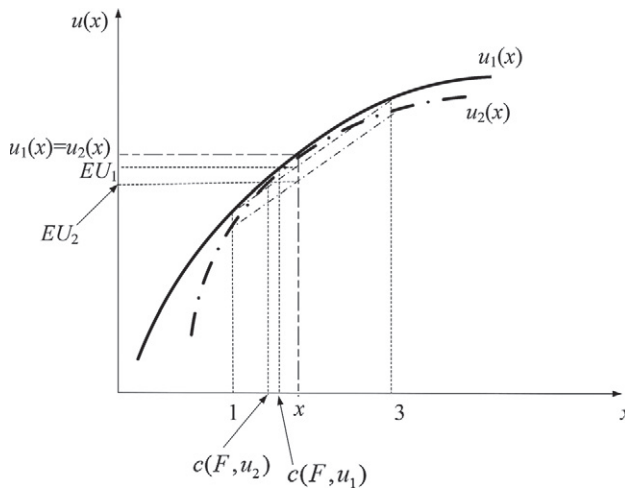


Figure 5.27
Different degrees of risk aversion

that of individual 1, $c(F, u_1)$, reflecting that individual 2 is more risk averse than individual 1.

5.8 Prudence

5.8.1 Absolute and Relative Prudence

As suggested in the previous section, risk aversion measures the concavity of an individual's utility function. An alternative interpretation of Arrow–Pratt's coefficient of relative risk aversion, $r_R(x)$, is that it takes the marginal utility of wealth, $u'(x)$, and examines its elasticity, $-xu''(x)/u'(x)$. A common criticism of this coefficient is that in standard applications in which risk enters linearly (i.e., with gains added to the initial wealth and losses subtracted), individuals do not necessarily choose to hold precautionary savings. More generally, the concept of risk aversion is not appropriate to explain individual behavior when risks affect marginal rather than total utility.²² In order to consider the effect of these risks, Kimball (1990) suggested the notion of “prudence” (as an individual's tendency to prepare for future risks), and provided two measures: absolute prudence,

$$K_A(x) = -\frac{u'''(x)}{u''(x)},$$

and relative prudence,

$$K_R(x) = -x \frac{u'''(x)}{u''(x)}.$$

Intuitively, if relative risk aversion weighs on the elasticity of marginal utility, relative prudence finds the elasticity of its concavity. In this context, Kimball (1990) regarded an individual as “prudent” if the third derivative of his utility is positive, $u'''(x) > 0$. Arrow–Pratt's coefficient of absolute risk aversion, $r_A(x)$, and the coefficient of relative prudence, $K_R(x)$, are closely related. In particular, differentiating $r_A(x)$ yields

$$r'_A(x) = -\frac{u''' \cdot u' - (u'' \cdot u'')}{(u')^2} = -\frac{u'''u' - (u'')^2}{(u')^2} - \frac{u'''}{u'} + \left(\frac{u''}{u'}\right)^2.$$

²² See Hahn (1970), Rothschild and Stiglitz (1971), Fishburn and Porter (1976), Cheung et al. (1987), Mitchell (1994), and the recent survey by Meyer and Meyer (2005).

Dividing both sides by $r_A(x) = -u''/u'$ yields

$$\frac{r'_A(x)}{r_A(x)} = -\frac{u'''}{u'} \left(-\frac{u'}{u''} \right) + \left(\frac{u''}{u'} \right)^2 \left(-\frac{u'}{u''} \right) = \frac{u'''}{u''} - \frac{u''}{u'} = -K_A(x) + r_A(x).$$

Hence

$$\frac{r'_A(x)}{r_A(x)} = -K_A(x) + r_A(x),$$

and solving for $K_A(x)$, we find an expression of absolute prudence as a function of $r_A(x)$,

$$K_A(x) = r_A(x) - \frac{r'_A(x)}{r_A(x)}.$$

Now, if we multiply both sides by x , we obtain

$$x \cdot K_A(x) = x \cdot r_A(x) - x \frac{r'_A(x)}{r_A(x)},$$

and since $K_R(x) \equiv x \cdot K_A(x)$ and $r_R(x) \equiv x \cdot r_A(x)$ by definition, we find an expression of relative prudence, $K_R(x)$, as a function of $r_R(x)$ and $r_A(x)$:

$$K_R(x) = r_R(x) - x \cdot \frac{r'_A(x)}{r_A(x)}.$$

Example 5.11: Measuring prudence in CARA and CRRA functions Let us evaluate the degree of relative prudence for the utility functions considered in examples 5.7 (CARA) and 5.8 (CRRA). In the case of CARA, $u(x) = -e^{-ax}$, where $a > 0$, we found $r_A(x) = a$ for all x and $r_R(x) = x \cdot a$, which entails a relative prudence of

$$K_R(x) = x \cdot a - x \cdot \frac{0}{x \cdot a} = x \cdot a = r_R(x)$$

implying that, in this case, relative prudence coincides with relative risk aversion. In the case of the CRRA in example 5.8, where $u(x) = x^b$, we found that $r_A(x) = (1-b)/x$ and $r_R(x) = 1-b$, thus yielding a relative prudence of

$$K_R(x) = (1-b) - x \cdot \frac{(1-b)/x^2}{(1-b)/x} = (1-b) = -b,$$

implying that, in this case, $K_R(x) < r_R(x)$. ■

5.8.2 Cautioness and Temperance

The literature has also analyzed the relationship between prudence and risk aversion that we explored in the examples above using the following ratio:

$$C(x) = \frac{K_R(x)}{r_R(x)}.$$

This ratio is often interpreted as measuring the “cautioness” of the individual, since a positive ratio $C(x) > 0$ implies $K_R(x) > r_R(x)$, indicating that the individual wants to hedge against the downside risk of an investment. As suggested in the previous examples, the CARA utility function yields $C(x) = 1$, whereas the CRRA entails $C(x) < 1$. The literature has also developed the coefficient of “temperance,” $T(x) = -u'''(x)/u''(x)$, where an individual is deemed as “temperate” when the fourth derivative of his utility function is negative, $u''''(x) \leq 0$, and describes his desire to totally reduce the exposure to risks. For a recent article unifying these approaches, see the review by Eeckhoudt (2012) and references therein.

5.9 Prospect Theory and Reference-Dependent Utility

5.9.1 Prospect Theory

Kahneman and Tversky (1979, 1986) proposed a model of choice under uncertainty often referred as “prospect theory,” in which a decision maker’s total value from a list of possible outcomes $x = (x_1, x_2, \dots, x_n)$ with associated probabilities $p = (p_1, p_2, \dots, p_n)$ is given by

$$v(x, p) = \sum_{i=1}^n w(p_i) v(x_i).$$

Hence prospect theory deviates from expected utility theory in three dimensions. First, probabilities are weighted with probability weighting function $w(p_i)$. At $w(p_i) = p_i$, this model would coincide with standard expected utility theory. However, at $w(p_i) > p_i$, the individuals overestimate the likelihood of outcome x_i , and at $w(p_i) < p_i$, they underestimate it. (We assume, however, that over- or underestimation does not arise in outcomes occurring with certainty, i.e., $w(0) = 0$ and $w(1) = 1$.) Second, payoffs are evaluated relative to a reference point x_0 with the so-called “value function” $v(x_i)$ describing the utility that this individual obtains from outcome x_i . As depicted in figure 5.28, this value function is increasing and concave, $v''(x_i) < 0$, for all outcomes that constitute a gain relative to reference point x_0 , $x_i > x_0$, meaning the decision maker is